KDIGO CLINICAL PRACTICE GUIDELINE FOR LIPID MANAGEMENT IN CHRONIC KIDNEY DISEASE

Supplemental Tables
November 2013

Supplemental Table 1: Summary table of RCT examining the effect of exercise in CKD 5HD patients [continuous outcomes]

Outcome (Units)	Author, Year Country Ref #	Duration Outcome measurement (Treatment)	Description		No. Analyzed (Enrolled)		Baseline		Results (Lipids)				P value	Quality
			Intervention	Control	Intervention	Control	GFR or S_{Cr}	Proteinuria	Baseline Intervention (Control)	Final Intervention (Control)	Δ Intervention (Control)	Net Δ (95% CI)		
Lipid levels														
ΔTotal cholesterol, mmol/L	van Vlsteren 2005 The Netherlands[1]	3 mo (3 mo)	Exercise program	Control	53 (60)	43 (43)	CKD 5HD	nd	4.6 (4.7)	4.6 (4.6)	0 (-0.1)	0.1	NS	Good

Supplemental Table 2: Summary table of RCT examining low vs. moderate protein diet in CKD patients without DM [categorical outcomes]

Outcome	Author, Year Country Ref #	Duration Outcome measurement (Treatment)	Description		No. Analyzed (Enrolled)		Baseline						Results		P value	Quality
			Intervention	Control	Intervention	Control	GFR	DM (%)	TC	LDL	HDL	Tg	Events No. (%) Intervention [Control]	RR/OR/HR (95% CI)[1]		
									mg/dL							
Mortality																
All-cause mortality	Cianciaruso 2009 Italy[2]	4 y (3 y)	Low-protein diet	Moderate-protein diet	212 (212)	211 (211)	16 (17) mL/min/ 1.73 m^2	11 (13)	nd	125 (124)	nd	nd	23 (11%) [25 (12%)]	HR 1.12 (0.64, 1.99)	nd	Fair
ESRD	Cianciaruso 2009 Italy[2]	4 y (3 y)	Low-protein diet	Moderate-protein diet	212 (212)	211 (211)	16 (17) mL/min/ 1.73 m^2	11 (13)	nd	125 (124)	nd	nd	42 (20) [41 (19)]	HR 1.00 (0.65, 1.55)	nd	Fair

[1] Adjusted for age, sex, comorbidity score index, and basal estimated glomerular filtration rate; as well as time-variant covariates including estimated glomerular filtration rate, protein intake, serum phosphate level, therapy with erythropoiesis-stimulating agents, and therapy with angiotensin-converting enzyme inhibitors, angiotensin receptor blockers, or both.

Supplemental Table 3: Summary table of RCT examining low vs. moderate protein diet in CKD patients without DM [continuous outcomes]

Outcome	Author, Year Country Ref #	Duration Outcome measurement (Treatment)	Description		No. Analyzed (Enrolled)		Baseline		Results (Lipids)				P value	Quality
			Intervention	Control	Intervention	Control	GFR	DM (%)	Baseline Intervention (Control)	Final Intervention (Control)	Δ Intervention (Control)	Net Δ (95% CI)		
Lipid levels														
ΔLDL, mg/dL	Cianciaruso 2009 Italy[2]	6 mo (3 y)	Low-protein diet	Moderate-protein diet	206 (212)	198 (211)	16 (17) mL/min/ 1.73 m^2	11 (13)	125 (124)	118 (122)	-7 (-2)	-5 (-13, 3)	NS	Fair
		1 y (3 y)			199 (212)	191 (211)				118 (120)	-7 (-4)	-3 (-10, 5)	NS	Fair
		2 y (3 y)			189 (212)	187 (211)				116 (115)	-9 (-9)	0 (-7, 7)	NS	Fair
		2 y (3 y)			181 (212)	178 (211)				111 (123)	-14 (-1)	-13 (-21, -5)	0.001	Fair
		3 y (3 y)			167 (212)	164 (211)				110 (121)	-15 (-3)	-12 (-20, -4)	0.002	Fair
		3 y (3 y)			146 (212)	147 (211)				118 (124)	-7 (0)	-7 (-15, 1)	NS (0.07)	Fair
		4 y (3 y)			132 (212)	139 (211)				112 (121)	-13 (3)	-10 (-17, 2)	0.012	Fair
		4 y (3 y)			127 (212)	124 (211)				113 (111)	-12 (-13)	1 (-7, 9)	NS	Fair

Supplemental Table 4: Summary table of RCT examining statin therapy vs. lifestyle modification in kidney transplant recipients without DM [categorical outcomes]

Outcome	Author, Year Country Ref #	Duration Outcome measurement (Treatment)	Description		No. Analyzed (Enrolled)		Baseline						Results		P value	Quality
			Intervention	Control	Intervention	Control	GFR or S_{Cr}	DM (%)	TC (mg/dL)	LDL (mg/dL)	HDL (mg/dL)	Tg (mg/dL)	Events No. (%) Intervention [Control]	RR/OR/HR (95% CI)		
CV events																
Focal cerebellar infarctions	Nart 2009 Turkey[3]	4 y (4 y)	Fluvastatin 40 mg	NCEP modified Step 1 diet	90 (90)	53 (53)	GFR 71.3 (69.7) mL/min S_{Cr} 1.32 (1.43) mg/dL	nd	231 (187)	135 (99)	63 (56)	170 (139)	2 (2%) [0 (0%)]	nd	nd	Fair

* Results are for adjusted analysis if provided. If unadjusted results are different please annotate.

Supplemental Table 5: Summary table of RCT examining statin therapy vs. lifestyle modification in kidney transplant recipients without DM [continuous outcomes]

Outcome	Author, Year Country Ref #	Duration Outcome measurement (Treatment)	Description		No. Analyzed (Enrolled)		Baseline		Results (Lipids)				P value	Quality
			Intervention	Control	Intervention	Control	GFR or S_{Cr}	DM (%)	Baseline Intervention (Control)	Final Intervention (Control)	Δ Intervention (Control)	Net Δ (95% CI)		
Lipid levels														
ΔLDL, mg/dL	Nart 2009 Turkey[3]	1 y (4 y)	Fluvastatin 40 mg	NCEP modified Step 1 diet	90 (90)	53 (53)	GFR 71.3 (69.7) mL/min S_{Cr} 1.32 (1.43) mg/dL	nd	135 (99.2)	115.6 (96.2)	-18.9 (-3.0)	-15.9	<0.000	Fair
		4 y (4 y)								123.4 (101.3)	-11.1 (+2.1)	-13.2	<0.000	Fair
ΔHDL, mg/dL		1 y (4 y)							63 (56)	53.2) (49.2)	-9.7 (-6.5)	-3.2	NS	Fair
		4 y (4 y)								48.9 (46.1)	-14.0 (-9.6)	-4.4	NS	Fair
ΔTriglycerides, mg/dL		1 y (4 y)							170 (139)	169.7 (143.4)	-0.6 (+4.7)	-5.3	0.035	Fair
		4 y (4 y)								163.0 (145.3)	-7.3 (+6.6)	-0.7	NS	Fair
ΔTotal cholesterol, mg/dL		1 y (4 y)							231.2 (187.3)	202.7 (172.7)	-28.5 (-14.6)	-13.9	0.00	Fair
		4 y (4 y)								206.0 (177.7)	-25.2 (-9.6)	-15.6	0.00	Fair

Supplemental Table 6: Summary table of RCT examining statin therapy vs. usual care in patients with CKD without DM [categorical outcomes]

Outcome	Author, Year Country Ref #	Duration Outcome measurement (Treatment)	Description		No. Analyzed (Enrolled)		Baseline						Results		P value	Quality
			Intervention	Control	Intervention	Control	GFR or S_{Cr}	DM (%)	TC (mg/dL)	LDL (mg/dL)	HDL (mg/dL)	Tg (mg/dL)	Events No. (%) Intervention [Control]	RR/OR/HR (95% CI)		
Composite outcome																
ESRD or Halving of GFR	ALLHAT (CKD subgp) Rahman 2008 Multi[4]	5 y (5 y)	Pravastatin 40 mg/d	Usual care	779 (779)	778 (778)	GFR 50.8 (50.6) mL/min/1.73 m^2	32 (30)	226 (224)	147 (145)	46 (46)[2]	165 (164)	52 (7) [50 (6)]	RR 0.97 (0.66, 1.43)	nd	Good[3]
ESRD or 25% decline in GFR													156 (20) [154 (20)]	RR 0.98 (0.79, 1.22)	nd	Good[4]
Kidney function																
ESRD	ALLHAT (CKD subgp) Rahman 2008 Multi[4]	5 y (5 y)	Pravastatin 40 mg/d	Usual care	779 (779)	778 (778)	GFR 50.8 (50.6) mL/min/1.73 m^2	nd	226 (224)	147 (145)	46 (46)[5]	165 (164)	32 (4) [31 (4)]	RR 1.05 (0.64, 1.73)	nd	Good[6]

* Results are for adjusted analysis if provided. If unadjusted results are different please annotate.

[2] Estimated from graph
[3] Study is graded "Good" however consider downgrading in EP due to the fact that "there was no significant interaction of baseline eGFR and treatment group".
[4] Study is graded "Good" however consider downgrading in EP due to the fact that "there was no significant interaction of baseline eGFR and treatment group".
[5] Estimated from graph
[6] Study is graded "Good" however consider downgrading in EP due to the fact that "there was no significant interaction of baseline eGFR and treatment group".

Supplemental Table 7: Summary table of RCT examining statin therapy vs. usual care in patients with CKD without DM [continuous outcomes]

Outcome	Author, Year Country Ref #	Duration Outcome measurement (Treatment)	Description		No. Analyzed (Enrolled)		Baseline		Results (Lipids)				P value	Quality
			Intervention	Control	Intervention	Control	GFR or S_{Cr}	DM (%)	Baseline Intervention (Control)	Final Intervention (Control)	Δ Intervention (Control)	Net Δ (95% CI)		
Lipid levels														
ΔTotal cholesterol, mg/dL	ALLHAT (CKD subgp) Rahman 2008 Multi[4]	5 y (5 y)	Pravastatin 40 mg/d	Usual care	779 (779)	778 (778)	GFR 50.8 (50.6) mL/min/1.73 m^2	32 (30)	226 (224)	178 (198)[7]	-48.2 (-26.1)	-22.1	nd	Good[8]
ΔHDL, mg/dL									46 (46)[9]	56 (38)[10]	+10 (-8)	18	nd	Good[11]

[7] Estimated from graph
[8] Study is graded "Good" however consider downgrading in EP due to the fact that "there was no significant interaction of baseline eGFR and treatment group".
[9] Estimated from graph
[10] Estimated from graph
[11] Study is graded "Good" however consider downgrading in EP due to the fact that "there was no significant interaction of baseline eGFR and treatment group".

Supplemental Table 8: Summary table of RCTs of statins vs. placebo in patients with CKD with and without DM [categorical outcomes]

Outcome	Author, Year Country Ref #	Duration Outcome measurement (Treatment)	Description		No. Analyzed (Enrolled)		Baseline						Results		P value	Quality
			Intervention	Control	Intervention	Control	GFR or S_{Cr}	DM (%)	TC	LDL	HDL	Tg	Events No. (%) Intervention [Control]	RR/OR/HR (95% CI)		
									mg/dL							
CKD with DM																
Composite outcomes																
Coronary heart disease death, nonfatal MI, CABG, or PTCA (primary)	CARE, LIPID, WOSCOPS (CKD subgp) Tonelli 2005 Multi[5]	5 y (5 y)	Pravastatin	Placebo	571 (571)		eGFR 58 ml/min/1.73 m^2 S_{Cr} 1.3 mg/dL	100 (100)	213	140	36	181	nd	HR 0.75[12] (0.57, 0.98)	<0.05	Fair
Coronary heart disease death, nonfatal MI, CABG, PTCA, or stroke														HR 0.79[13] (0.62, 1.03)	NS (0.06)	Fair
Coronary heart disease death or nonfatal MI														HR 0.84[14] (0.60, 1.18)	NS	Fair
Mortality																
All-cause mortality	CARE, LIPID, WOSCOPS (CKD subgp) Tonelli 2005 Multi[5]	5 y (5 y)	Pravastatin	Placebo	571 (571)		eGFR 58 ml/min/1.73 m^2 S_{Cr} 1.3 mg/dL	100 (100)	213	140	36	181	nd	HR 0.98[15] (0.69, 1.39)	NS	Fair

[12] Hazards ratio have been adjusted for age; SBP; HDL; LDL; triglyceride; an indicator for trial (CARE, LIPID, or WOSCOPS); current smoking status; history of stroke; history of coronary disease; insulin dependence; and baseline use of aspirin, β blockers, ACEi, and CCB
[13] Hazards ratio have been adjusted for age; SBP; HDL; LDL; triglyceride; an indicator for trial (CARE, LIPID, or WOSCOPS); current smoking status; history of stroke; history of coronary disease; insulin dependence; and baseline use of aspirin, β blockers, ACEi, and CCB
[14] Hazards ratio have been adjusted for age; SBP; HDL; LDL; triglyceride; an indicator for trial (CARE, LIPID, or WOSCOPS); current smoking status; history of stroke; history of coronary disease; insulin dependence; and baseline use of aspirin, β blockers, ACEi, and CCB
[15] Hazards ratio have been adjusted for age; SBP; HDL; LDL; triglyceride; an indicator for trial (CARE, LIPID, or WOSCOPS); current smoking status; history of stroke; history of coronary disease; insulin dependence; and baseline use of aspirin, β blockers, ACEi, and CCB

Outcome	Author, Year Country Ref #	Duration Outcome measurement (Treatment)	Description		No. Analyzed (Enrolled)		Baseline						Results		P value	Quality
			Intervention	Control	Intervention	Control	GFR or S_{Cr}	DM (%)	TC	LDL	HDL	Tg	Events No. (%) Intervention [Control]	RR/OR/HR (95% CI)		
									mg/dL							
All cause mortality	CARDS (CKD subgp) Colhoun, 2009 UK Ireland[6]	4 y (4 y)	Atorvastatin	Placebo	482 (482)	488 (488)	eGFR 53.5 (54.1) mL/min/1.73 m²	100 (100)	210 (212)	120 (120)	56 (56)	≤600 (≤600)	27 (6%) [30 (6%)]	HR 0.86[16] (0.51, 1.45)	NS	Good
CV events																
CABG or PTCA	CARE, LIPID, WOSCOPS (CKD subgp) Tonelli 2005 Multi[5]	5 y (5 y)	Pravastatin	Placebo	571 (571)		eGFR 58 ml/min/1.73 m² S_{Cr} 1.3 mg/dL	100 (100)	213	140	36	181	nd	HR 0.69[17] (0.47, 1.01)	NS (0.06)	Fair
Any stroke														HR 1.12[18] (0.63,1.97)	NS	Fair
Major CV disease	CARDS (CKD subgp) Calhoun, 2009 UK Ireland[6]	4 y (4 y)	Atorvastatin	Placebo	482 (482)	488 (488)	eGFR 53.5 (54.1) mL/min/1.73 m²	100 (100)	210 (212)	120 (120)	56 (56)	≤600 (≤600)	25 (5%) [42 (9%)]	HR 0.57[19] (0.35, 0.94)	0.02	Good
Stroke													6 (1%) [15 (3%)]	HR 0.38[20] (0.15, 0.99)	0.04	Good
Coronary heart disease													18 (4%) [27 (6%)]	HR 0.65[21] (0.36, 1.17)	NS	Good
Coronary revascularization													5 (1%) [12 (2%)]	HR 0.40[22] (0.14, 1.15)	NS (0.07)	Good
CKD without DM																
Composite outcomes																

16 Adjusted for age, sex and treatment group
17 Hazards ratio have been adjusted for age; SBP; HDL; LDL; triglyceride; an indicator for trial (CARE, LIPID, or WOSCOPS); current smoking status; history of stroke; history of coronary disease; insulin dependence; and baseline use of aspirin, β blockers, ACEi, and CCB
18 Hazards ratio have been adjusted for age; SBP; HDL; LDL; triglyceride; an indicator for trial (CARE, LIPID, or WOSCOPS); current smoking status; history of stroke; history of coronary disease; insulin dependence; and baseline use of aspirin, β blockers, ACEi, and CCB
19 Adjusted for age, sex and treatment group
20 Adjusted for age, sex and treatment group
21 Adjusted for age, sex and treatment group
22 Adjusted for age, sex and treatment group

Outcome	Author, Year Country Ref #	Duration Outcome measurement (Treatment)	Description		No. Analyzed (Enrolled)		Baseline						Results		P value	Quality
			Intervention	Control	Intervention	Control	GFR or S_{Cr}	DM (%)	TC	LDL	HDL	Tg	Events No. (%) Intervention [Control]	RR/OR/HR (95% CI)		
									mg/dL							
Nonfatal MI, nonfatal stroke, hospital stay for unstable angina, arterial revascularization, or confirmed CV death (primary)	JUPITER Ridker 2010 Multi[7]	2 y (2 y)	Rosuvastatin	Placebo	1638 (1638)	1629 (1629)	eGFR 56 ml/min/1.73 m^2	0 (0)	189	109	49	130	40 (2%) [71 (4%)]	HR 0.55 (0.38, 0.82)	0.002	Good
MI, stroke, or confirmed CV death													24 (2%) [40 (3%)]	HR 0.59 (0.36, 0.99)	0.04	Good
Nonfatal MI, nonfatal stroke, hospital stay for unstable angina, arterial revascularization, or confirmed CV death and all-cause mortality													64 (4%) [114 (7%)]	HR 0.55 (0.41, 0.75)	0.0001	Good
Nonfatal MI, nonfatal stroke, hospital stay for unstable angina, arterial revascularization, or confirmed CV death and all-cause mortality and venous thromboembolism													69 (4%) [127 (8%)]	HR 0.53 (0.40, 0.71)	<0.0001	Good

Outcome	Author, Year Country Ref #	Duration Outcome measurement (Treatment)	Description		No. Analyzed (Enrolled)		Baseline						Results		P value	Quality
			Intervention	Control	Intervention	Control	GFR or S_{Cr}	DM (%)	TC	LDL	HDL	Tg	Events No. (%) Intervention [Control]	RR/OR/HR (95% CI)		
									mg/dL							
Coronary heart disease death, nonfatal MI, CABG, or PTCA (primary)	CARE, LIPID, WOSCOPS (CKD subgp) Tonelli 2005 Multi[5]	Median 5 y (5 y)	Pravastatin	Placebo	4099 (4099)		eGFR 56.5 ml/min/1.73 m^2 S_{Cr} 1.3 mg/dL	0 (0)	224	153	38	161	nd	HR 0.77[23] (0.68, 0.87)	<0.05	Fair
Coronary heart disease death, nonfatal MI, CABG, PTCA, or stroke														HR 0.80[24] (0.71, 0.90)	<0.05	Fair
Coronary heart disease death or nonfatal MI														HR 0.85[25] (0.73, 1.00)	0.05	Fair
First primary CV event including cardiac death, nonfatal MI, resuscitated cardiac arrest, cardiac revascularization or unstable angina requiring hospitalization	ALLIANCE (CKD subgp) Koren 2009 US[8]	5 y (5 y)	Atorvastatin	Usual care	286 (286)	293 (293)	eGFR 51.3 (51.1) mL/min/1.73 m^2	30 (26)	228 (227)	148 (146)	40 (40)	200 (207)	78 (27%) [105 (36%)]	HR 0.72 (0.54, 0.97)	0.02	Fair
Nonfatal MI or cardiac death													32 (11%) [54 (18%)]	HR 0.55 (0.35, 0.85)	0.008	Fair

[23] Hazards ratio have been adjusted for age; SBP; HDL; LDL; triglyceride; an indicator for trial (CARE, LIPID, or WOSCOPS); current smoking status; history of stroke; history of coronary disease; insulin dependence; and baseline use of aspirin, β blockers, ACEi, and CCB
[24] Hazards ratio have been adjusted for age; SBP; HDL; LDL; triglyceride; an indicator for trial (CARE, LIPID, or WOSCOPS); current smoking status; history of stroke; history of coronary disease; insulin dependence; and baseline use of aspirin, β blockers, ACEi, and CCB
[25] Hazards ratio have been adjusted for age; SBP; HDL; LDL; triglyceride; an indicator for trial (CARE, LIPID, or WOSCOPS); current smoking status; history of stroke; history of coronary disease; insulin dependence; and baseline use of aspirin, β blockers, ACEi, and CCB

Outcome	Author, Year Country Ref #	Duration Outcome measurement (Treatment)	Description		No. Analyzed (Enrolled)		Baseline						Results		P value	Quality
			Intervention	Control	Intervention	Control	GFR or S_{Cr}	DM (%)	TC	LDL	HDL	Tg	Events No. (%) Intervention [Control]	RR/OR/HR (95% CI)		
									mg/dL							
CV mortality and hospitalization for CV morbidity (primary)	PREVEND IT Asselbergs 2004 Netherlands[9]	4 y (4 y)	Pravastatin	Placebo	433 (433)	431 (431)	S_{Cr} 91 (90) μmol/L	3 (2)	5.8 (5.8) mmol/L	4.1 (4.0) mmol/L	1.0 (1.0) mmol/L	1.4 (1.3) mmol/L	22 (5%) [25 (6%)]	HR 0.87 (0.49, 1.57)	NS	Fair
Cardiac death or MI	LIPS (CKD subgp) Lemos 2005 Multi[10]	4 y (4 y)	Fluvastatin	Placebo	150 (150)	160 (160)	S_{Cr} 1.33 mg/dL CrCl <47 mL/min	Total 12	200	131	39	150	7 (5%) [13 (8%)]	RR 0.57 (0.24, 1.40)	NS	Fair[26]
All cause death or MI													7 (5%) [13 (8%)]	RR 0.57 (0.24, 1.40)	NS	Fair[27]
Unstable angina, fatal and nonfatal MI, and sudden death	AFCAPS/TexCAPS (CKD subgp) Kendrick 2010 US[11]	5 y (5 y)	Lovastatin	Placebo	145 (145)	159 (159)	GFR 53 (53) mL/min/1.73 m^2 S_{Cr} 1.4 (1.4) mg/dL	1 (2)	224 (220)	151 (151)	39 (39)	177 (168)	5 (3.4%) 17 (7.5%)	RR 0.32 (0.10, 1.11)	NS (0.06)	Good[28]
Mortality																
All-cause mortality	JUPITER Ridker 2010 Multi[7]	2 y (2 y)	Rosuvastatin	Placebo	1638 (1638)	1629 (1629)	eGFR 56 ml/min/1.73 m^2	0 (0)	189	109	49	130	34 (2%) [61 (4%)]	HR 0.56 (0.37, 0.85)	0.005	Good

[26] Study graded "Fair" due to unbalance in baseline characteristics. No info on treatment vs. placebo among the 310 renal impairment patients. Consider downgrading further in EP due to being unable to locate interaction results.

[27] Study graded "Fair" due to unbalance in baseline characteristics. No info on treatment vs. placebo among the 310 renal impairment patients. Consider downgrading further in EP due to being unable to locate interaction results.

[28] Study is graded "Good", however consider downgrading in EP due to the fact that for all clinical events, the effects of lovastatin did not differ significantly between subgroups with and without CKD(P>0.1) for all interaction tests.

Outcome	Author, Year Country Ref #	Duration Outcome measurement (Treatment)	Description		No. Analyzed (Enrolled)		Baseline						Results		P value	Quality
			Intervention	Control	Intervention	Control	GFR or S_{Cr}	DM (%)	TC	LDL	HDL	Tg	Events No. (%) Intervention [Control]	RR/OR/HR (95% CI)		
									mg/dL							
All-cause mortality	CARE, LIPID, WOSCOPS (CKD subgp) Tonelli 2005 Multi[5]	5 y (5 y)	Pravastain	Placebo	4099 (4099)		eGFR 56.5 ml/min/1.73 m^2 S_{Cr} 1.3 mg/dL	0 (0)	224	153	38	161	nd	HR 0.97[29] (0.82, 1.15)	NS	Fair
All-cause mortality	ALLIANCE (CKD subgp) Koren 2009 US[8]	5 y (5 y)	Atorvastatin	Usual care	286 (286)	293 (293)	eGFR 51.3 (51.1) mL/min/1.73 m^2	30 (26)	228 (227)	148 (146)	40 (40)	200 (207)	47 (16%) [59 (20%)]	RR 0.82 (0.58, 1.15)[30]	NS	Fair
All-cause mortality	PREVEND IT Asselbergs 2005 Netherlands[9]	4 y (4 y)	Pravastatin	Placebo	433 (433)	431 (431)	S_{Cr} 91 (90) µmol/L	3 (2)	5.8 (5.8) mmol/L	4.1 (4.0) mmol/L	1.0 (1.0) mmol/L	1.4 (1.3) mmol/L	6 (1%) [4 (1%)]	RR 1.49 (0.42-5.25)[31]	NS	Fair
All-cause mortality	LIPS (CKD subgp) Lemos 2005 Multi[10]	4 y (4 y)	Fluvastatin	Placebo	150 (150)	160 (160)	S_{Cr} 1.33 mg/dL CrCl <47 mL/min	Total 12	200	131	39	150	3 (2%) [3 (2%)]	RR 1.07 (0.22, 5.20)	NS	Fair[32]
Noncardiac death													0 (0%) [0 (0%)]	--	nd	Fair[33]

[29] Hazards ratio have been adjusted for age; SBP; HDL; LDL; triglyceride; an indicator for trial (CARE, LIPID, or WOSCOPS); current smoking status; history of stroke; history of coronary disease; insulin dependence; and baseline use of aspirin, β blockers, ACEi, and CCB

[30] Calculated by the ERT

[32] Study graded “Fair” due to unbalance in baseline characteristics. No info on treatment vs. placebo among the 310 renal impairment patients. Consider downgrading further in EP due to being unable to locate interaction results.

[33] Study graded “fair” due to unbalance in baseline characteristics. No info on treatment vs. placebo among the 310 renal impairment patients. Consider downgrading further in EP due to being unable to locate interaction results.

Outcome	Author, Year Country Ref #	Duration Outcome measurement (Treatment)	Description		No. Analyzed (Enrolled)		Baseline						Results		P value	Quality
			Intervention	Control	Intervention	Control	GFR or S_{Cr}	DM (%)	TC (mg/dL)	LDL (mg/dL)	HDL (mg/dL)	Tg (mg/dL)	Events No. (%) Intervention [Control]	RR/OR/HR (95% CI)		
All-cause mortality (primary)	4S (CKD subgp) Chonchol 2007 Scandinavian countries[12]	5 y (5 y)	Simvastatin	Placebo	1143 (1143)	1171 (1171)	GFR 65.2 (65.2) mL/min)[34] SCr 1.15 (1.14) mg/dL	5 (4)	261 (261)	189 (189)	46 (46)	131 (134)	Total 246 (11%)	HR 0.69 (0.54, 0.89)	nd	Good[35]
Total mortality	MEGA (CKD subgp) Nakamura 2009 Japan[13]	5 y (5 y)	Pravastatin + NCEP Step 1 Diet	NCEP Step 1 Diet	1462 (1462)	1516 (1516)	GFR 52.6 (52.5) mL/min/m^2	Total 19	6.3 mmol/L	4.0 mmol/L	1.5 mmol/L	1.5 mmol/L	16 (1%) [34 (2%)]	HR 0.49 (0.27, 0.89)	0.02	Fair[36]
CV Mortality																
Death from CHD or nonfatal MI (primary)	CARE (CKD subgp) Tonelli 2003 US & Canada[14]	5 y (5 y)	Pravastatin	Placebo	844 (844)	867 (867)	CrCl 61.2 (61.3) mL/min[37] Scr 61.2 (61.3) µmol/L	13 (15)	209 (209)	138 (139)	41 (41)	149 (149)	89 (11%) [126 (15%)]	HR 0.72[38] (0.55, 0.95)	0.02	Good
Total mortality													86 (10%) [111 (13%)]	HR 0.81[39] (0.61, 1.08)	NS	Good
Cardiac death	LIPS (CKD subgp) Lemos 2005 Multi[10]	4 y (4 y)	Fluvastatin	Placebo	150 (150)	160 (160)	S_{Cr} 1.33 mg/dL CrCl CrCl <47 mL/min	Total 12	200	131	39	150	3 (2%) [3 (2%)]	RR 1.07 (0.22, 5.20)	NS	Fair[40]

[34] Mild chronic renal insufficiency is defined as eGFR <75 mL/min/1.73m^2 (<1.25 mL/s) or creatinine clearance <75 mL/min (1.25 mL/s)
[35] Study is graded "Good" however consider downgrading in EP due to inconsistency in interaction test results (P=0.05 for total mortality and P=0.84 for major coronary events).
[36] Study is graded "Fair" due to the fact that baseline characteristics by intervention were not provided in the CKD subgroup. Consider downgrading again in EP due to not being able to find interaction results.
[37] Mild chronic renal insufficiency is defined as creatinine clearance <75 mL/min
[38] Adjusted for age; sex; history of HTN; smoking at baseline; DM; previous CHF; use of ACEi, CCB, β-adrenergic blockers, and aspirin; proteinuria; SBP and DBP; baseline HDL and LDL, cholesterol levels; baseline Tg; serum albumin levels; BSA; and pravastatin use.
[39] Adjusted for age; sex; history of HTN; smoking at baseline; DM; previous CHF; use of ACEi, CCB, β-adrenergic blockers, and aspirin; proteinuria; SBP and DBP; baseline HDL and LDL, cholesterol levels; baseline Tg; serum albumin levels; BSA; and pravastatin use.
[40] Study graded "Fair" due to unbalance in baseline characteristics. No info on treatment vs. placebo among the 310 renal impairment patients. Consider downgrading further in EP due to being unable to locate interaction results.

Outcome	Author, Year Country Ref #	Duration Outcome measurement (Treatment)	Description		No. Analyzed (Enrolled)		Baseline						Results		P value	Quality
			Intervention	Control	Intervention	Control	GFR or S_{Cr}	DM (%)	TC	LDL	HDL	Tg	Events No. (%) Intervention [Control]	RR/OR/HR (95% CI)		
									mg/dL							
CV mortality	AFCAPS/TexCAPS (CKD subgp) Kendrick 2010 US[11]	5 y (5 y)	Lovastatin	Placebo	145 (145)	159 (159)	GFR 53 (53) mL/min/1.73m^2 S_{Cr} 1.4 (1.4) mg/dL	1 (2)	224 (220)	151 (151)	39 (39)	177 (168)	0 (0%) [1 (0.6%)]	--	nd	Good[41]
CV mortality	PREVEND IT Asselbergs 2005 Netherlands[9]	4 y (4 y)	Pravastatin	Placebo	433 (433)	431 (431)	S_{Cr} 91 (90) μmol/L	3 (2)	5.8 (5.8) mmol/L	4.1 (4.0) mmol/L	1.0 (1.0) mmol/L	1.4 (1.3) mmol/L	4 (1%) [4 (1%)]	nd	nd	Fair
CV events																
MI	JUPITER Ridker 2010 Multi[7]	2 y (2 y)	Rosuvastatin	Placebo	1638 (1638)	1629 (1629)	eGFR 56 ml/min/1.73 m^2	0 (0)	189	109	49	130	8 (1%) [20 (1%)]	HR 0.40 (0.17, 0.90)	0.02	Good
Stroke													10 (1%) [14 (1%)]	HR 0.71 (0.31, 1.59)	NS	Good
Arterial revascularization													19 (1%) [39 (1%)]	HR 0.48 (0.28, 0.83)	0.006	Good
Venous thromboembolism													6 (0.3%) [17 (1%)]	HR 0.34 (0.14, 0.88)	0.02	Good
Major coronary event	CARE Tonelli 2003 US & Canada[14]	5 y (5 y)	Pravastatin	Placebo	844 (844)	867 (867)	CrCl 61.2 (61.3) mL/min[42] Scr 61.2 (61.3)	13 (15)	209 (209)	138 (139)	41 (41)	149 (149)	171 (20%) [234 (27%)]	HR 0.72[43] (0.59, 0.88)	0.001	Good
Fatal MI or confirmed nonfatal MI													65 (8%) [90 (10%)]	HR 0.73[44] (0.52, 1.01)	NS (0.06)	Good

[41] Study is graded "Good", however consider downgrading in EP due to the fact that for all clinical events, the effects of lovastatin did not differ significantly between subgroups with and without CKD(P>0.1) for all interaction tests.

[42] Mild chronic renal insufficiency is defined as creatinine clearance <75 mL/min

[43] Adjusted for age; sex; history of HTN; smoking at baseline; DM; previous CHF; use of ACEi, CCB, β-adrenergic blockers, and aspirin; proteinuria; SBP and DBP; baseline HDL and LDL, cholesterol levels; baseline Tg; serum albumin levels; BSA; and pravastatin use.

[44] Adjusted for age; sex; history of HTN; smoking at baseline; DM; previous CHF; use of ACEi, CCB, β-adrenergic blockers, and aspirin; proteinuria; SBP and DBP; baseline HDL and LDL, cholesterol levels; baseline Tg; serum albumin levels; BSA; and pravastatin use.

Outcome	Author, Year Country Ref #	Duration Outcome measurement (Treatment)	Description: Intervention	Description: Control	No. Analyzed (Enrolled): Intervention	No. Analyzed (Enrolled): Control	Baseline: GFR or S_{Cr} μmol/L	Baseline: DM (%)	Baseline: TC (mg/dL)	Baseline: LDL (mg/dL)	Baseline: HDL (mg/dL)	Baseline: Tg (mg/dL)	Results: Events No. (%) Intervention [Control]	Results: RR/OR/HR (95% CI)	P value	Quality
CABG or PTCA													105 (12%) [153 (18%)]	HR 0.65[45] (0.50, 0.83)	0.001	Good
Unstable angina													133 (16%) [142 (16%)]	HR 0.93[46] (0.73, 1.18)	NS	Good
Stroke													29 (3%) [46 (5%)]	HR 0.62[47] (0.39, 1.00)	0.051	Good
Major coronary event in subgroup of CKD and proteinuria					255 (255)	268 (268)	nd	Dipstick +					65 (26%) [81 (30%)]	Unadjusted absolute reduction in cumulative incidence 4.7	nd	Good
Major coronary event in subgroup with CrCl >75 mL/min					1139 (1139)	1164 (1164)	CrCl >75 mL/min	nd	nd	nd	nd	nd	244 (21%) [315 (27%)]	Unadjusted absolute reduction in cumulative incidence 5.7	nd	Good
Major coronary event in subgroup with CrCl ≤75 mL/min					844 (844)	867 (867)	CrCl ≤75 mL/min						171 (20%) [234 (27%)]	Unadjusted absolute reduction in cumulative incidence 6.7	nd	Good

45 Adjusted for age; sex; history of HTN; smoking at baseline; DM; previous CHF; use of ACEi, CCB, β-adrenergic blockers, and aspirin; proteinuria; SBP and DBP; baseline HDL and LDL, cholesterol levels; baseline Tg; serum albumin levels; BSA; and pravastatin use.
46 Adjusted for age; sex; history of HTN; smoking at baseline; DM; previous CHF; use of ACEi, CCB, β-adrenergic blockers, and aspirin; proteinuria; SBP and DBP; baseline HDL and LDL, cholesterol levels; baseline Tg; serum albumin levels; BSA; and pravastatin use.
47 Adjusted for age; sex; history of HTN; smoking at baseline; DM; previous CHF; use of ACEi, CCB, β-adrenergic blockers, and aspirin; proteinuria; SBP and DBP; baseline HDL and LDL, cholesterol levels; baseline Tg; serum albumin levels; BSA; and pravastatin use.

Outcome	Author, Year Country Ref #	Duration Outcome measurement (Treatment)	Description: Intervention	Description: Control	No. Analyzed (Enrolled): Intervention	No. Analyzed (Enrolled): Control	Baseline: GFR or S_{Cr}	Baseline: DM (%)	Baseline: TC (mg/dL)	Baseline: LDL (mg/dL)	Baseline: HDL (mg/dL)	Baseline: Tg (mg/dL)	Results: Events No. (%) Intervention [Control]	Results: RR/OR/HR (95% CI)	P value	Quality
Major coronary event in subgroup with CrCl 60-75 mL/min					524 (524)	518 (518)	CrCl 60-75 mL/min						103 (20%) [143 (28%)]	Unadjusted absolute reduction in cumulative incidence 7.9	nd	Good
Major coronary event in subgroup with CrCl 50-75 mL/min					719 (719)	736 (736)	CrCl 50-75 mL/min						140 (20%) [207 (28%)]	Unadjusted absolute reduction in cumulative incidence 8.6	nd	Good
Major coronary event in subgroup with CrCl ≤60 mL/min					320 (320)	349 (349)	CrCl ≤60 mL/min						68 (21%) [91 (26%)]	Unadjusted absolute reduction in cumulative incidence 4.8	nd	Good
Major coronary event in subgroup with CrCl ≤50 mL/min					125 (125)	131 (131)	CrCl ≤50 mL/min						31 (25%) [27 (21%)]	--	nd	Good
CABG or PTCA	CARE, LIPID,	5 y (5 y)	Pravastain	Placebo	4099 (4099)		eGFR 56.5	0 (0)	224	153	38	161	nd	HR 0.72[48] (0.61, 0.86)	<0.05	Fair

[48] Hazards ratio have been adjusted for age; SBP; HDL; LDL; triglyceride; an indicator for trial (CARE, LIPID, or WOSCOPS); current smoking status; history of stroke; history of coronary disease; insulin dependence; and baseline use of aspirin, β blockers, ACEi, and CCB

Outcome	Author, Year Country Ref #	Duration Outcome measurement (Treatment)	Description		No. Analyzed (Enrolled)		Baseline						Results		P value	Quality
			Intervention	Control	Intervention	Control	GFR or S_{Cr}	DM (%)	TC	LDL	HDL	Tg	Events No. (%) Intervention [Control]	RR/OR/HR (95% CI)		
									mg/dL							
Any stroke	WOSCOPS (CKD subgp) Tonelli 2005 Multi[5]						ml/min/1.73 m^2 S_{Cr} 1.3 mg/dL							HR 0.96[49] (0.71, 1.30)	NS	Fair
Non-fatal MI	ALLIANCE (CKD subgp) Koren 2009 US[8]	5 y (5 y)	Atorvastatin	Usual care	286 (286)	293 (293)	eGFR 51.3 (51.1) mL/min/1.73 m^2	30 (26)	228 (227)	148 (146)	40 (40)	200 (207)	17 (6%) [29 (10%)]	HR 0.54 (0.30, 0.99)	NS	Fair
Stroke													11 (4%) [12 (4%)]	RR 0.94 (0.42, 2.09)[50]	NS	Fair
Cardiac death													17 (6%) [27 (9%)]	RR 0.65 (0.36,1.16)[51]	NS	Fair
Fatal and non fatal CV events	AFCAPS/TexCAPS (CKD subgp) Kendrick 2010 US[11]	5 y (5 y)	Lovastatin	Placebo	145 (145)	159 (159)	GFR 53 (53) mL/min/1.73 m^2 S_{Cr} 1.4 (1.4) mg/dL	1 (2)	224 (220)	151 (151)	39 (39)	177 (168)	8 (6%) [21 (13%)]	RR 0.39 (0.16, 0.93)	0.03[52]	Good[53]
Fatal and non fatal MI													2 (1%) [6 (4%)]	RR 0.10 (0.01, 1.32)	NS (0.08)	Good[54]
Major coronary events	4S (CKD subgp) Chonchol	5 y (5 y)	Simvastatin	Placebo	1143 (1143)	1171 (1171)	GFR 65.2 (65.2) mL/min)[55]	5 (4)	261 (261)	189 (189)	46 (46)	131 (134)	Total (24%)	HR 0.67 (0.56, 0.79)	nd	Good[56]

[49] Hazards ratio have been adjusted for age; SBP; HDL; LDL; triglyceride; an indicator for trial (CARE, LIPID, or WOSCOPS); current smoking status; history of stroke; history of coronary disease; insulin dependence; and baseline use of aspirin, β blockers, ACEi, and CCB

[50] Calculated by the ERT

[51] Calculated by the ERT

[52] Fully adjusted model

[53] Study is graded "Good", however consider downgrading in EP due to the fact that for all clinical events, the effects of lovastatin did not differ significantly between subgroups with and without CKD(P>0.1) for all interaction tests.

[54] Study is graded "Good", however consider downgrading in EP due to the fact that for all clinical events, the effects of lovastatin did not differ significantly between subgroups with and without CKD(P>0.1) for all interaction tests.

[55] Mild chronic renal insufficiency is defined as eGFR <75 mL/min/1.73m² (<1.25 mL/s) or creatinine clearance <75 mL/min (1.25 mL/s)

[56] Study is graded "Good" however consider downgrading in EP due to inconsistency in interaction test results (P=0.05 for total mortality and P=0.84 for major coronary events).

Outcome	Author, Year Country Ref #	Duration Outcome measurement (Treatment)	Description		No. Analyzed (Enrolled)		Baseline						Results		P value	Quality
			Intervention	Control	Intervention	Control	GFR or S_{Cr}	DM (%)	TC	LDL	HDL	Tg	Events No. (%) Intervention [Control]	RR/OR/HR (95% CI)		
									mg/dL							
Nonfatal MI	2007 Scandinavian countries[12]						SCr 1.15 (1.14) mg/dL						nd	HR 0.65 (0.55, 0.78)	nd	Good[57]
Coronary revascularization													nd	HR 0.62 (0.49, 0.77)	nd	Good[58]
CV events	PREVEND IT Asselbergs 2005 Netherlands[9]	4 y (4 y)	Pravastatin	Placebo	433 (433)	431 (431)	S_{Cr} 91 (90) μmol/L	3 (2)	5.8 (5.8) mmol/L	4.1 (4.0) mmol/L	1.0 (1.0) mmol/L	1.4 (1.3) mmol/L	18 (4%) [21 (5%)]	nd	nd	Fair
Stroke	MEGA (CKD subgp) Nakamura 2009 Japan[13]	5 y (5 y)	Pravastatin + NCEP Step 1 Diet	NCEP Step 1 Diet	1462 (1462)	1516 (1516)	GFR 52.6 (52.5) mL/min/m^2	Total 19	6.3 mmol/L	4.0 mmol/L	1.5 mmol/L	1.5 mmol/L	8 (1%) [29 (2%)]	HR 0.27 (0.12, 0.59)	0.01	Fair[59]
Kidney function																
Doubling of S_{Cr}	JUPITER Ridker 2010 Multi[7]	2 y (2 y)	Rosuvastatin	Placebo	1638 (1638)	1629 (1629)	eGFR 56 ml/min/1.73 m^2	0 (0)	189	109	49	130	3 (0.2%) [0 (0%)]	nd	nd	Good
↓≥25% MDRD-eGFR in patients with mild CKD	CARE, LIPID, WOSCOPS (CKD subgp) Tonelli 2005 Multi[15]	Median 5 y (5 y)	Pravastain	Placebo	6479 (6479)	6364 (6364)	eGFR 73.8 (73.8) mL/min/1.73 m^2 S_{Cr} 1.08 (1.08) mg/dL	6 (6)	236 (236)	163 (163)	40 (40)	160 (158)	nd (10%) [nd (11%)]	RR 0.94 (0.85, 1.03)	NS	Fair
Acute renal failure in patients with mild CKD													nd (0.2%) [nd [0.5%)]	RR 0.42 (0.22, 0.78)	0.006	Fair

[57] Study is graded "Good" however consider downgrading in EP due to inconsistency in interaction test results (P=0.05 for total mortality and P=0.84 for major coronary events).
[58] Study is graded "Good" however consider downgrading in EP due to inconsistency in interaction test results (P=0.05 for total mortality and P=0.84 for major coronary events).
[59] Study is graded "Fair" due to the fact that baseline characteristics by intervention were not provided in the CKD subgroup. Consider downgrading again in EP due to not being able to find interaction results.

Outcome	Author, Year Country Ref #	Duration Outcome measurement (Treatment)	Description		No. Analyzed (Enrolled)		Baseline						Results		P value	Quality
			Intervention	Control	Intervention	Control	GFR or S_{Cr}	DM (%)	TC	LDL	HDL	Tg	Events No. (%) Intervention [Control]	RR/OR/HR (95% CI)		
									mg/dL							
eGFR ≤60 ml/min/1.73m^2 in patients with mild CKD													nd (27%) [nd [29%)]	RR 0.95 (0.90, 1.00)	NS (0.06)	Fair
↓≥25% CG-eGFR in patients with mild CKD					<6479 (<6479)	<6364 (<6364)	nd	6 (6)	nd	nd	nd	nd	nd (9%) [nd (10%)]	RR 0.88 (0.78, 0.91)	0.03	Fair
↓eGFR ≥25%	AFCAPS/TexCAPS (CKD subgp) Kendrick 2010 US[11]	5 y (5 y)	Lovastatin	Placebo	145 (145)	159 (159)	GFR 53 (53) mL/min/1.73 m^2 S_{Cr} 1.4 (1.4) mg/dL	1 (2)	224 (220)	151 (151)	39 (39)	177 (168)	4% [3%]	nd	NS	Good[60]
↓GFR≥25%	4S (CKD subgp) Huskey 2009 Scandinavian countries[16]	6 y (6 y)	Simvastatin	Placebo	199 (199)	210 (210)	GFR 55 (55) mL/min/1.73 m^2[61]	4 (1)	265 (265)	192 (191)	47 (48)	133 (134)	5 (3%) [13 (6%)]	OR 0.21 (0.05, 0.94)	0.04	Good[62]

[60] Study is graded "Good", however consider downgrading in EP due to the fact that for all clinical events, the effects of lovastatin did not differ significantly between subgroups with and without CKD(P>0.1) for all interaction tests.
[61] Mild chronic renal insufficiency is defined as eGFR <60 mL/min/1.73m^2
[62] Study is graded "Good" however consider downgrading in EP due to inconsistency in interaction test results (P=0.05 for total mortality and P=0.84 for major coronary events).

Supplemental Table 9: Summary table of RCTs of statins vs. placebo in various stages of CKD with and without DM [continuous outcomes]

Outcome (Units)	Author, Year Country Ref #	Duration Outcome measurement (Treatment)	Description		No. Analyzed (Enrolled)		Baseline		Results (Lipids)				P value	Quality
			Intervention	Control	Intervention	Control	GFR or S_{Cr}	DM (%)	Baseline Intervention (Control)	Final Intervention (Control)	Δ Intervention (Control)	Net Δ (95% CI)		
CKD without DM														
Lipid levels														
Median LDL, mg/dL	JUPITER Ridker 2010 Multi[7]	2 y (2 y)	Rosuvastatin	Placebo	1638 (1638)	1629 (1629)	eGFR 56 ml/min/1.73 m²	0 (0)	189 (189)	55 (108)	-134 (-81)	-53	<0.001	Good
Median HDL, mg/dL									49 (49)	53 (50)	4 (1)	3	nd	Good
Median Triglycerides, mg/dL									130 (130)	99 (129)	-31 (-1)	-30	nd	Good
ΔLDL, mg/dL	CARE Tonelli 2003[17] US & Canada	5 y (5 y)	Pravastatin	Placebo	345 (345)	345 (345)	GFR 53.2 (52.5) mL/min/1.73 m² S_{Cr} 1.4 (1.4) mg/dL	14 (17)	139.5 (138.7)	nd	-41 (nd)	nd	nd	Fair
Total cholesterol, mg/dL									210.0 (209.9)	nd	-41 (nd)	nd	nd	Fair
%ΔTotal cholesterol, mg/dL	ALLIANCE (CKD subgp) Koren 2009 US[8]	5 y (5 y)	Atorvastatin	Usual care	271 (286)	158 (293)	eGFR 51.3 (51.1) mL/min/1.73 m²	30 (26)	228.4 (227.0)	nd	-24 (-15)[63]	-9	<0.001	Fair
ΔLDL, mg/dL									148.2 (146.0)	92.2 (106.1)	-34.5 (-24.2)	-10.3	<0.001	Fair
%ΔHDL , mg/dL									40.2 (40.3)	nd	+4 (+8)[64]	4	0.1	Fair
%ΔTriglycerides, mg/dL									200.3 (207.3)	nd	-8 (-4)[65]	-4	0.5	Fair
ΔTotal cholesterol, mmol/L	PREVEND IT Asselbergs 2004 Netherlands[9]	4 y (4 y)	Pravastatin	Placebo	376 (433)	382 (431)	S_{Cr} 91 (90) µmol/L	3 (2)	5.8 (5.8) mmol/L	4.8 (5.6) mmol/L	-1 (-0.2)	-0.8	<0.05[66]	Fair
ΔLDL cholesterol, mmol/L					375 (433)	379 (431)			4.1 (4.0) mmol/L	3.1 (3.9) mmol/L	-1 (-0.1)	-0.9	<0.05[67]	Fair

[63] Estimated from figure
[64] Estimated from figure
[65] Estimated from figure
[66] Results of Δ in total cholesterol from baseline to 3 months, 1 year, 2 years and 3 years was also statistically significant.
[67] Results of Δ in LDL from baseline to 3 months, 1 year, 2 years and 3 years was also statistically significant.

Outcome (Units)	Author, Year Country Ref #	Duration Outcome measurement (Treatment)	Description		No. Analyzed (Enrolled)		Baseline		Results (Lipids)				P value	Quality
			Intervention	Control	Intervention	Control	GFR or S_{Cr}	DM (%)	Baseline Intervention (Control)	Final Intervention (Control)	Δ Intervention (Control)	Net Δ (95% CI)		
ΔTotal cholesterol, mg/dL	AFCAPS/TexCAPS (CKD subgp) Kendrick 2010 US[11]	5 y (5 y)	Lovastatin	Placebo	145 (145)	159 (159)	GFR 53 (53) mL/min/ 1.73 m^2 S_{Cr} 1.4 (1.4) mg/dL	1 (2)	224 (220)	nd	-20% (+1.5%)	-18.5%	nd	Good[68]
ΔHDL, mg/dL									39 (39)	nd	+7.4% (+1.1%)	+6.3%	nd	Good[69]
ΔLDL, mg/dL									151 (151)	nd	-27% (+1.9%)	-25.1%	nd	Good[70]
ΔTriglycerides, mg/dL									177 (168)	nd	-15% (+4.2%)	-10.8%	nd	Good[71]
ΔTotal cholesterol, mg/dL	4S (CKD subgp) Chonchol 2007 Scandinavian countries[12]	5 y (5 y)	Simvastatin	Placebo	1143 (1143)	1171 (1171)	GFR 65.2 (65.2) mL/min)[72] SCr 1.15 (1.14) mg/dL	5 (4)	261 (261)	233 (261)	-28% (0%)	-28%	nd	Good[73]
ΔHDL, mg/dL									46 (46)	52 (43)	+6% (-3%)	+3%	nd	Good[74]
ΔLDL, mg/dL									189 (189)	151 (186)	-38% (+2%)	-36%	nd	Good[75]
ΔTriglycerides, mg/dL									131 (134)	115 (136)	-16% (+2%)	-14%	nd	Good[76]

68 Study is graded "Good", however consider downgrading in EP due to the fact that for all clinical events, the effects of lovastatin did not differ significantly between subgroups with and without CKD(P>0.1) for all interaction tests.
69 Study is graded "Good", however consider downgrading in EP due to the fact that for all clinical events, the effects of lovastatin did not differ significantly between subgroups with and without CKD(P>0.1) for all interaction tests.
70 Study is graded "Good", however consider downgrading in EP due to the fact that for all clinical events, the effects of lovastatin did not differ significantly between subgroups with and without CKD(P>0.1) for all interaction tests.
71 Study is graded "Good", however consider downgrading in EP due to the fact that for all clinical events, the effects of lovastatin did not differ significantly between subgroups with and without CKD(P>0.1) for all interaction tests.
72 Mild chronic renal insufficiency is defined as eGFR <75 mL/min/1.73m^2 (<1.25 mL/s) or creatinine clearance <75 mL/min (1.25 mL/s)
73 Study is graded "Good" however consider downgrading in EP due to inconsistency in interaction test results (P=0.05 for total mortality and P=0.84 for major coronary events).
74 Study is graded "Good" however consider downgrading in EP due to inconsistency in interaction test results (P=0.05 for total mortality and P=0.84 for major coronary events).
75 Study is graded "Good" however consider downgrading in EP due to inconsistency in interaction test results (P=0.05 for total mortality and P=0.84 for major coronary events).
76 Study is graded "Good" however consider downgrading in EP due to inconsistency in interaction test results (P=0.05 for total mortality and P=0.84 for major coronary events).

Supplemental Table 10: Evidence profile of RCTs examining the effect of statins vs. placebo in patients with CKD with and without DM

Outcome		# of studies and study design	Total N (treatment)	Methodological quality of studies per outcome	Consistency across studies	Directness of the evidence generalizability/ applicability	Other considerations	Summary of findings: Quality of evidence for outcome	Qualitative description of effect	Importance of outcome
Composite outcomes	DM	MA of 3 RCTs (High)	571 (nd)	Some limitations (-1)	NA	Uncertainty about directness (-1)	None (0)	Low	Possible benefit from statins in patients with DM	Critical
	Non DM	MA of 3 RCTs + 5 RCTs (High)	9423 (2652 + nd)	No limitations (0)	No important inconsistencies (0)	Uncertainty about directness (-1)	None (0)	Moderate	Benefit from statins in patients without DM	
Mortality	DM	MA of 3 RCTs + 1 RCT (High)	1541 (482 + nd)	No limitations (0)	No important inconsistencies (0)	Uncertainty about directness (-1)	None (0)	Moderate	No difference in patients with DM	Critical
	Non DM	MA of 3 RCTs + 6 RCTs (High)	14411 (5112 + nd)	No limitations (0)	Important inconsistencies (-1)	Uncertainty about directness (-1)	None (0)	Low	Possible benefit from statins in patients without DM	
CV mortality	DM	0 RCTs	--	--	--	--	--	--	--	Critical
	Non DM	4 RCTs (High)	3186 (1572)	No limitations (0)	Important inconsistencies (-1)	Uncertainty about directness (-1)	None (0)	Low	Possible benefit from statins in patients without DM	
CV events	DM	MA of 3 RCTs + 1 RCT (High)	1541 (482 + nd)	No limitations (0)	No important inconsistencies (0)	Uncertainty about directness (-1)	None (0)	Moderate	Possible benefit from statins for patients with DM	Critical
	Non DM	MA of 3 RCTs + 7 RCTs (High)	16116 (5951 + nd)	No limitations (0)	No important inconsistencies (0)	Uncertainty about directness (-1)	None (0)	Moderate	Benefit from statins in patients without DM	
ESRD	DM	0 RCTs	--	--	--	--	--	--	--	Critical
	Non DM	0 RCTs	--	--	--	--	--	--		
Kidney function (categorical)	DM	0 RCTs	--	--	--	--	--	--	--	High
	Non DM	MA of 3 RCTs + 3 RCTs (High)	18728 (9405)	Some limitations (-1)	No important inconsistencies (0)	Uncertainty about directness (-1)	None (0)	Low	Possible benefit from statins in patients without DM	
Lipid levels (continuous)	DM	0 RCTs	--	--	--	--	--	--	--	Moderate
	Non DM	6 RCTs (High)	7762 (3918)	No limitations (0)	No important inconsistencies (0)	Uncertainty about directness (-1)	None (0)	Moderate	Benefit from statins in patients without DM	
Adverse events		MA of 3 RCTs + 8 RCTs (High)	16256 (5951 + nd)						No difference	Moderate
Total		MA of 3 RCTs + 8 RCTs (High)	16256 (5951 + nd)							

Outcome	# of studies and study design	Total N (treatment)	Methodological quality of studies per outcome	Consistency across studies	Directness of the evidence generalizability/ applicability	Other considerations	Summary of findings		
							Quality of evidence for outcome	Qualitative description of effect	Importance of outcome
Balance of potential benefits and harms: Possible benefit							**Quality of overall evidence:** Low		

Supplemental Table 11: Summary table of RCTs of statins vs. placebo in dialysis patients with and without DM [categorical outcomes]

Outcome	Author, Year Country Ref #	Duration Outcome measurement (Treatment)	Description: Intervention	Description: Control	No. Analyzed (Enrolled): Intervention	No. Analyzed (Enrolled): Control	Baseline: GFR or S_{Cr}	Baseline: DM (%)	Baseline: TC (mg/dL)	Baseline: LDL (mg/dL)	Baseline: HDL (mg/dL)	Baseline: Tg (mg/dL)	Results: Events No. (%) Intervention [Control]	Results: RR/OR/HR (95% CI)	P value	Quality
Dialysis with DM																
Composite outcomes																
Composite of cardiac death, nonfatal MI, or stroke (primary)	4D Wanner 2005 Germany[18]	4 y (2 y)	Atorvastatin	Placebo	619 (619)	636 (636)	CKD 5HD	100 (100)	218 (220)	125 (127)	36 (36)	261 (267)	226 (37%) [243 (38%)]	RR 0.92 (0.77, 1.10)	NS	Good
Composite of cardiac death, nonfatal MI, or stroke for CRP quartile 1 (≤2.3 mg/L)	4D Krane 2008 Germany[19]	4 y (2 y)			316 (316)				nd	126	nd	nd	60 [50]	HR 1.19 (0.81,1.76)	NS	Fair
Composite of cardiac death, nonfatal MI, or stroke for CRP quartile 2 (>2.3-≤5 mg/L)					310 (310)					128			50 [56]	HR 0.75 (0.50, 1.10)	NS	Fair
Composite of cardiac death, nonfatal MI, or stroke for CRP quartile 3 (>5-≤12.4 mg/L)					312 (312)					126			52 [78]	HR 0.79 (0.55, 1.13)	NS	Fair
Composite of cardiac death, nonfatal MI, or stroke for CRP quartile 4 (>12.4 mg/L)					311 (311)					122			62 [57]	HR 1.06 (0.73, 1.54)	NS	Fair
Primary outcome in 2nd and 3rd CRP quartiles combined					622 (622)					nd			nd	HR 0.85 (0.65, 1.10)	NS	Fair

Outcome	Author, Year Country Ref #	Duration Outcome measurement (Treatment)	Description		No. Analyzed (Enrolled)		Baseline						Results		P value	Quality
			Intervention	Control	Intervention	Control	GFR or S_{Cr}	DM (%)	TC mg/dL	LDL mg/dL	HDL mg/dL	Tg mg/dL	Events No. (%) Intervention [Control]	RR/OR/HR (95% CI)		
Major CV event, defined as nonfatal MI, nonfatal stroke, or death from CV causes	AURORA Fellstrom subgp with DM 2009 Multi[20]	Median 4 y (2 y)	Rosuvastatin	Placebo	388 (388)	343 (343)	CKD 5HD	100 (100)	nd	nd	nd	nd	135 (13%) [136 (15%)]	RR 0.88 (0.73, 1.06)[77]	nd	Fair
Composite cardiac endpoint of cardiac death or nonfatal MI	AURORA Holdaas subgp with DM 2011[21] Multi	3 y (2 y)	Rosuvastatin	Placebo	388 (388	343 (343)	CKD 5HD	100%	4.49 (4.35) mmol/L	2.51 (2.43) mmol/L	1.11 (1.08) mmol/L	1.90 (1.85) mmol/L	85 (22%) [104 (30%)]	HR 0.68 (0.51, 0.90)	0.008	Good
Cardiac death, nonfatal MA, fatal or nonfatal stroke													Risk reduction 16.2% for DM	HR 0.838 (0.654, 1.074)	NS	Good
Mortality																
Death from all causes	4D Wanner 2005 Germany[18]	4 y (2 y)			619 (619)	636 (636)			218 (220)	125 (127)	36 (36)	261 (267)	297 (48) [320 (50%)]	RR 0.93 (0.79, 1.08)	NS	Good
Death from causes other than CV or cerebrovascular disease													149 (24%) [158 (25%)]	RR 0.95 (0.76, 1.18)	NS	Good
All-cause mortality for CRP quartile 1 (≤2.3 mg/L)	4D Krane 2008 Germany[19]	4 y (2 y)	Atorvastatin	Placebo	316 (316)		CKD 5HD	100 (100)		126			64 [55]	HR 1.07 (0.74, 1.55)	NS	Fair
All-cause mortality for CRP quartile 2 (>2.3-≤5 mg/L)					310 (310)				nd	128	nd	nd	67 [77]	HR 0.73 (0.53, 1.02)	NS (0.067)	Fair
All-cause mortality for CRP quartile 3 (>5-≤12.4 mg/L)					312 (312)					126			63 [94]	HR 0.79 (0.57, 1.11)	NS	Fair

[77] Calculated by ERT

Outcome	Author, Year Country Ref #	Duration Outcome measurement (Treatment)	Description		No. Analyzed (Enrolled)		Baseline						Results		P value	Quality
			Intervention	Control	Intervention	Control	GFR or S_{Cr}	DM (%)	TC	LDL	HDL	Tg	Events No. (%) Intervention [Control]	RR/OR/HR (95% CI)		
									mg/dL							
All-cause mortality for CRP quartile 4 (>12.4 mg/L)					311 (311)					122			101 [91]	HR 1.02 (0.77, 1.37)	NS	Fair
All-cause mortality in 2nd and 3rd CRP quartiles combined					622 (622)					nd			130 [171]	HR 0.78 (0.62, 0.99)	0.038	Fair
Death from any cause	AURORA Holdaas subgp with DM 2011[21] Multi	3 y (2 y)	Rosuvastatin	Placebo	388 (388)	343 (343)	CKD 5HD	100%	4.49 (4.35) mmol/L	2.51 (2.43) mmol/L	1.11 (1.08) mmol/L	1.90 (1.85) mmol/L	219 (56%) [213 (62%)]	HR 0.86 (0.71, 1.04)	NS	Good
CV Mortality																
Cardiac mortality	4D Wanner 2005 Germany [18]	4 y (2 y)	Atorvastatin	Placebo	619 (619)	636 (636)	CKD 5HD	100 (100)	218 (220)	125 (127)	36 (36)	261 (267)	121 (20%) [149 (23%)]	RR 0.81 (0.64, 1.03)	NS (0.08)	Good
Deaths attributable to cardiac disease	AURORA Holdaas subgp with DM 2011[21] Multi	3 y (2 y)	Rosuvastatin	Placebo	388 (388)	343 (343)	CKD 5HD	100%	4.49 (4.35) mmol/L	2.51 (2.43) mmol/L	1.11 (1.08) mmol/L	1.90 (1.85) mmol/L	35% [47%]	--	nd	Good
Fatal cardiac events													64% [71%]	--	nd	Good
CV events																
Nonfatal MI													70 (11%) [79 (12%)]	RR 0.88 (0.64, 1.21)	NS	Good
Fatal stroke													27 (4%) [13 (2%)]	RR 2.03 (1.05, 3.93)	0.04	Good
Nonfatal stroke													33 (5%) [32 (5%)]	RR 1.04 (0.64, 1.69)	NS	Good
All cardiac events combined	4D Wanner 2005 Germany[18]	4 y (2 y)	Atorvastatin	Placebo	619 (619)	636 (636)	CKD 5HD	100 (100)	218 (220)	125 (127)	36 (36)	261 (267)	205 (33%) [246 (39%)]	RR 0.82 (0.68, 0.99)	0.03	Good
All cerebrovascular events combined													79 (13%) [70 (11%)]	RR 1.12 (0.81, 1.55)	NS	Good
Stroke													59 (10%) [44 (7%)]	RR 1.33 (0.90, 1.97)	NS	Good

Outcome	Author, Year Country Ref #	Duration Outcome measurement (Treatment)	Description		No. Analyzed (Enrolled)		Baseline						Results		P value	Quality
			Intervention	Control	Intervention	Control	GFR or S_{Cr}	DM (%)	TC (mg/dL)	LDL (mg/dL)	HDL (mg/dL)	Tg (mg/dL)	Events No. (%) Intervention [Control]	RR/OR/HR (95% CI)		
Hemorrhagic stroke	AURORA Fellstrom subgp with DM 2009 Multi[20]	Median 4 y (2 y)	Rosuvastatin	Placebo	388 (388)	343 (343)	CKD 5HD	100 (100)	nd	nd	nd	nd	12 (4%) [2 (1%)]	RR 5.30 (1.20-23.53)[78]	0.07	Fair
Stroke	AURORA Holdaas subgp with DM 2011[21] Multi	3 y (2 y)	Rosuvastatin	Placebo	388 (388	343 (343)	CKD 5HD	100%	4.49 (4.35) mmol/L	2.51 (2.43) mmol/L	1.11 (1.08) mmol/L	1.90 (1.85) mmol/L	38 (10%) [20 (6%)]	HR 1.65 (0.96, 2.83)	NS (0.07)	Good
Fatal stroke													18 (5%) [11 (3%)]	HR 1.41 (0.67, 2.99)	NS	Good
Hemorrhagic strokes													12 (3%) [2 (0.6%)]	HR 5.21 (1.17, 23.27)	0.031	Good
Dialysis without DM																
Composite outcome																
Time to CV event defined as a nonfatal MI, nonfatal stroke, or death from CV cause (primary)	AURORA Fellstrom 2009 Multi[20]	Median 4 y (2 y)	Rosuvastatin	Placebo	1389 (1391)	1384 (1385)	CKD 5HD	21 (18)	176 (174)	100 (99)	45 (45)	157 (154)	9.2/100 pt-y [9.5/100 pt-y]	--	--	Good
CV event defined as a nonfatal MI, nonfatal stroke, or death from CV cause (primary)													396 (29%) [408 (29%)]	HR 0.96 (0.84, 1.11)	NS	Good
Mortality																
Time to death from any cause	AURORA Fellstrom 2009 Multi[20]	Median 4 y (2 y)	Rosuvastatin	Placebo	1389 (1391)	1384 (1385)	CKD 5HD	21 (18)	176 (174)	100 (99)	45 (45)	157 (154)	13.5/100 pt-y [14.0/100 pt-y]	--	--	Good
Death from any cause													636 (46%) [660 (48%)]	HR 0.96 (0.86, 1.07)	NS	Good

[78] Calculated by ERT

Outcome	Author, Year Country Ref #	Duration Outcome measurement (Treatment)	Description		No. Analyzed (Enrolled)		Baseline						Results		P value	Quality
			Intervention	Control	Intervention	Control	GFR or S_{Cr}	DM (%)	TC (mg/dL)	LDL (mg/dL)	HDL (mg/dL)	Tg (mg/dL)	Events No. (%) Intervention [Control]	RR/OR/HR (95% CI)		
Time to death from non-CV cause													5.5/100 pt-y [6.0/100 pt-y]	--	--	Good
Death from non-CV cause													248 (18%) [268 (19%)]	HR 0.92 (0.77, 1.09)	NS	Good
CV mortality																
Time to CV mortality	AURORA Fellstrom 2009 Multi[20]	Median 4 y (2 y)	Rosuvastatin	Placebo	1389 (1391)	1384 (1385)	CKD 5HD	21 (18)	176 (174)	100 (99)	45 (45)	157 (154)	7.2/100 pt-y [7.3/100 pt-y]	--	--	Good
CV mortality													324 (23%) [324 (23%)]	HR 1.00 (0.85, 1.16)	NS	Good
CV events																
Time to nonfatal MI	AURORA Fellstrom 2009 Multi[20]	Median 4 y (2 y)	Rosuvastatin	Placebo	1389 (1391)	1384 (1385)	CKD 5HD	21 (18)	176 (174)	100 (99)	45 (45)	157 (154)	2.1/100 pt-y [2.5/100 pt-y]	--	--	Good
Nonfatal MI													91 (7%) [107 (8%)]	HR 0.84 (0.64, 1.11)	NS	Good
Time to nonfatal stroke													1.2/100 pt-y [1.1/100 pt-y]	--	--	Good
Nonfatal stroke													53 (4%) [45 (3%)]	HR 1.17 (0.79, 1.75)	NS	Good

Supplemental Table 12: Summary table of RCTs of statins vs. placebo in dialysis patients with and without DM [continuous outcomes]

Outcome (Units)	Author, Year Country Ref #	Duration Outcome measurement (Treatment)	Description		No. Analyzed (Enrolled)		Baseline		Results (Lipids)				P value	Quality
			Intervention	Control	Intervention	Control	GFR or S_{Cr}	DM (%)	Baseline Intervention (Control)	Final Intervention (Control)	Δ Intervention (Control)	Net Δ (95% CI)		
Dialysis with DM														
Lipid levels														
ΔLDL, mg/dL	4D Wanner 2005 Germany[18]	4 wk (2 y)	Atorvastatin	Placebo	619 (619)	636 (636)	CKD 5HD	100 (100)	125 (127)	72 (120)	-53 (-7)	-46	nd	Fair
		5 y (2 y)			44 (619)	37 (636)				70[79] (92)	-55 (-35)	-20	nd	Fair
ΔLDL, mmo/L	AURORA Holdaas subgp with DM 2011[21] Multi	3 mo (2 y)	Rosuvastatin	Placebo	388 (388	343 (343)	CKD 5HD	100	2.51 (2.43)	1.41 (2.43)	-1.1 (0)	-1.1	nd	Good
ΔTotal cholesterol, mmol/L									4.49 (4.35)	3.29 (nd)	-1.2 (nd)	nd	nd	Fair
Dialysis without DM														
Lipid levels														
ΔLDL, mg/dL	AURORA Fellstrom 2009 Multi[20]	Median 4y (2 y)	Rosuvastatin	Placebo	1389 (1391)	1384 (1385)	CKD 5HD	21 (18)	100 (99)	58 (97.1)	-42 (-1.9)	-43.9	<0.001	Good
ΔTotal cholesterol, mg/dL									176 (174)	129 (173)	-47 (-1)	-48	<0.001	Good
ΔTriglycerides, mg/dL									157 (154)	131 (155.8)	-26 (+1.8)	-24.2	<0.001	Good
ΔHDL, mg/dL									45 (45)	46.2 (44.6)	+1.2 (+0.4)	1.6	0.045	Good
ΔLDL, mg/dL									147 (145)	98 (135)[80]	-48.5 (-9.5)	-39	nd	Good[81]
ΔTriglycerides, mg/dL									165 (164)	129 (165)[82]	-35.5 (+1)	-36.5	nd	Good[83]

[79] Estimated from figure
[80] Estimated from graph
[81] Study is graded "Good" however consider downgrading in EP due to the fact that "there was no significant interaction of baseline eGFR and treatment group".
[82] Estimated from graph
[83] Study is graded "Good" however consider downgrading in EP due to the fact that "there was no significant interaction of baseline eGFR and treatment group".

Supplemental Table 13: Evidence profile of RCTs examining the effect of statins vs. placebo in dialysis patients with and without DM

<table>
<tr><th rowspan="2" colspan="2">Outcome</th><th rowspan="2"># of studies and study design</th><th rowspan="2">Total N (treatment)</th><th rowspan="2">Methodological quality of studies per outcome</th><th rowspan="2">Consistency across studies</th><th rowspan="2">Directness of the evidence generalizability/ applicability</th><th rowspan="2">Other considerations</th><th colspan="3">Summary of findings</th></tr>
<tr><th>Quality of evidence for outcome</th><th>Qualitative and quantitative description of effect</th><th>Importance of outcome</th></tr>
<tr><td rowspan="2">Composite outcomes</td><td>DM</td><td>2 RCTs (High)</td><td>1986 (1007)</td><td>No limitations (0)</td><td>No important inconsistencies (0)</td><td>Direct (0)</td><td>None (0)</td><td>High</td><td>No difference in patients with DM</td><td rowspan="2">Critical</td></tr>
<tr><td>Non DM</td><td>1 RCT (High</td><td>2773 (1389)</td><td>No limitations (0)</td><td>NA</td><td>Direct (0)</td><td>Sparse (-1)</td><td>Moderate</td><td>No difference in patients without DM</td></tr>
<tr><td rowspan="2">Mortality</td><td>DM</td><td>2 RCTs (High)</td><td>1986 (1007)</td><td>No limitations (0)</td><td>No important inconsistencies (0)</td><td>Direct (0)</td><td>Sparse (-1)</td><td>Moderate</td><td>No difference in patients with DM</td><td rowspan="2">Critical</td></tr>
<tr><td>Non DM</td><td>1 RCT (High</td><td>2773 (1389)</td><td>No limitations (0)</td><td>NA</td><td>Direct (0)</td><td>Sparse (-1)</td><td>Moderate</td><td>No difference in patients without DM</td></tr>
<tr><td rowspan="2">CV mortality</td><td>DM</td><td>2 RCTs (High)</td><td>1986 (1007)</td><td>No limitations (0)</td><td>No important inconsistencies (0)</td><td>Direct (0)</td><td>Sparse (-1)</td><td>Moderate</td><td>Possible benefit from statins in patients with DM</td><td rowspan="2">Critical</td></tr>
<tr><td>Non DM</td><td>1 RCT (High</td><td>2773 (1389)</td><td>No limitations (0)</td><td>NA</td><td>Direct (0)</td><td>Sparse (-1)</td><td>Moderate</td><td>No difference in patients without DM</td></tr>
<tr><td rowspan="2">CV events</td><td>DM</td><td>2 RCTs (High)</td><td>1790 (905)</td><td>No limitations (0)</td><td>No important inconsistencies (0)</td><td>Direct (0)</td><td>None (0)</td><td>Moderate</td><td>Possible benefit from statins for patients with DM</td><td rowspan="2">Critical</td></tr>
<tr><td>Non DM</td><td>1 RCT (High</td><td>2773 (1389)</td><td>No limitations (0)</td><td>NA</td><td>Direct (0)</td><td>Sparse (-1)</td><td>Moderate</td><td>No difference in patients without DM</td></tr>
<tr><td rowspan="2">ESRD</td><td>DM</td><td>0 RCTs</td><td>--</td><td>--</td><td>--</td><td>--</td><td>--</td><td>--</td><td rowspan="2">--</td><td rowspan="2">Critical</td></tr>
<tr><td>Non DM</td><td>0 RCTs</td><td>--</td><td>--</td><td>--</td><td>--</td><td>--</td><td>--</td></tr>
<tr><td rowspan="2">Kidney function (categorical)</td><td>DM</td><td>0 RCTs</td><td>--</td><td>--</td><td>--</td><td>--</td><td>--</td><td>--</td><td rowspan="2">--</td><td rowspan="2">High</td></tr>
<tr><td>Non DM</td><td>0 RCTs</td><td>--</td><td>--</td><td>--</td><td>--</td><td>--</td><td>--</td></tr>
<tr><td rowspan="2">Lipid levels (continuous)</td><td>DM</td><td>2 RCTs (High)</td><td>1986 (1007)</td><td>Some limitations (-1)</td><td>No important inconsistencies (0)</td><td>Uncertainty about directness (-1)</td><td>None (0)</td><td>Low</td><td>Benefit from statins in patients with DM</td><td rowspan="2">Moderate</td></tr>
<tr><td>Non DM</td><td>1 RCT (High</td><td>2773 (1389)</td><td>No limitations (0)</td><td>NA</td><td>Uncertainty about directness (-1)</td><td>Sparse (-1)</td><td>Low</td><td>Benefit from statins in patients without DM</td></tr>
<tr><td colspan="2">Adverse events</td><td>3 RCTs</td><td>4759 (2396)</td><td></td><td></td><td></td><td></td><td></td><td>No difference in AEs for patients with and without DM</td><td>Moderate</td></tr>
<tr><td colspan="2">Total</td><td>3 RCTs</td><td>4759 (2396)</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td colspan="8">Balance of potential benefits and harms:
No difference</td><td colspan="3">Quality of overall evidence:
Moderate</td></tr>
</table>

Supplemental Table 14: Summary table of RCT examining statin vs. placebo in patients with ADPKD [continuous outcomes]

Outcome	Author, Year Country Ref #	Duration Outcome measurement (Treatment)	Description		No. Analyzed (Enrolled)		Baseline		Results (Lipids)				P value	Quality
			Intervention	Control	Intervention	Control	GFR or S_{Cr}	DM (%)	Baseline Intervention (Control)	Final Intervention (Control)	Δ Intervention (Control)	Net Δ (95% CI)		
Lipid levels														
ΔTotal cholesterol, mmol/L	Fasset 2010 Australia[22]	2 y (2 y)	Pravastatin 20 mg/day	Control	29 (31)	20 (29)	GFR 58.5 (49.9) mL/min/1.73 m^2	nd	5.22 (4.91)	4.91 (4.95)	-0.35 (0.04)	-0.39	nd	Fair
ΔLDL, mmol/L									3.52 (3.08)	2.98 (2.92)	-0.38 (-0.16)	-0.22	nd	Fair
ΔHDL, mmol/L									1.08 (1.38)	1.29 (1.50)	0.10 (0.11)	-0.01	nd	Fair
ΔTriglycerides, mmol/L									1.65 (1.19)	1.47 (1.30)	-0.29 (0.11)	-0.4	nd	Fair

Supplemental Table 15: Summary table of RCT examining simvastatin/ezetimibe combination vs. simvastatin/placebo in CKD patients without DM [categorical outcomes]

Outcome	Author, Year Country Ref #	Duration Outcome measurement (Treatment)	Description		No. Analyzed (Enrolled)		Baseline						Results		P value	Quality
			Intervention	Control	Intervention	Control	GFR or S_{Cr}	DM (%)	TC (mg/dL)	LDL (mg/dL)	HDL (mg/dL)	Tg (mg/dL)	Events No. (%) Intervention [Control]	RR/OR/HR (95% CI)		
Mortality																
Death	UK-HARP II Landray 2006 UK[23]	6 mo (6 mo)	Simvastatin 20 mg/d Ezetimibe 10mg/d	Simvastatin 20 mg/d	102 (102)	101 (101)	GFR 26.1 (29.7) mL/min/1.73 m^2	12 (10)	198 (195)	121 (117)	40 (40)	167 (188)	3 (2%) [0 (0%)]	nd	nd	Good
Kidney function																
ESRD	UK-HARP II Landray 2006 UK[23]	6 mo (6 mo)	Simvastatin 20 mg/d Ezetimibe 10mg/d	Simvastatin 20 mg/d	102 (102)	101 (101)	GFR 26.1 (29.7) mL/min/1.73 m^2	12 (10)	198 (195)	121 (117)	40 (40)	167 (188)	14 (14%) [14 (14%)]	RR 0.99 (0.50, 1.97)[84]	NS	Good

[84] Calculated by ERT

Supplemental Table 16: Summary table of RCT examining simvastatin/ezetimibe combination vs. simvastatin/placebo in CKD patients without DM [continuous outcomes]

Outcome	Author, Year Country Ref #	Duration Outcome measurement (Treatment)	Description		No. Analyzed (Enrolled)		Baseline		Results (Lipids)				P value	Quality
			Intervention	Control	Intervention	Control	GFR or S_{Cr}	DM (%)	Baseline Intervention (Control)	Final Intervention (Control)	Δ Intervention (Control)	Net Δ (95% CI)		
Lipid levels														
ΔTotal cholesterol, mg/dL	UK-HARP II Landray 2006 UK[23]	6 mo (6 mo)	Simvastatin 20 mg/d Ezetimibe 10mg/d	Simvastatin 20 mg/d	102 (102)	101 (101)	GFR 26.1 (29.7) mL/min/1.73 m^2	12 (10)	198 (195)	139 (158)	-59 (-37)	-22[85]	<0.001	Good
ΔLDL, mg/dL									121 (117)	72 (86)	-49 (-31)	-18[86]	<0.001	Good
ΔHDL, mg/dL									40 (40)	40 (39)	0 (-1)	1[87]	NS	Good
ΔTriglycerides, mg/dL									167 (188)	135 (171)	-32 (-17)	-16[88]	NS	Good

85 Mean absolute difference adjusted for baseline differences
86 Mean absolute difference adjusted for baseline differences
87 Mean absolute difference adjusted for baseline differences
88 Mean absolute difference adjusted for baseline differences

Supplemental Table 17: Summary table of RCT of statin + ezetimibe vs. placebo in CKD patients [categorical outcomes]

Outcome	Author, Year Country Ref #	Duration Outcome measurement (Treatment)	Description		No. Analyzed (Enrolled)		Baseline						Results		P value	Quality
			Intervention	Control	Intervention	Control	GFR or SCr (units)	DM (%)	TC	LDL	HDL	Tg	Events No. (%) Intervention [Control]	RR/OR/HR (95% CI)		
									(mmol/L)							
Composite outcomes																
ESRD or death	SHARP 2011 Multi[24]	5 y (4 y)	Simvastatin 20 mg/d + ezetimibe 10 mg/d	Placebo	4650 (4650)	4620 (4620)	GFR 27 (27) mL/min/ 1.73 m^2	23 (23)	4.88 (4.90)	2.77 (2.78)	1.12 (1.11)	2.31 (2.34)	1477 (47%) [1513 (48%)]	RR 0.97 (0.90, 1.04)	NS	Good
Mortality																
All-cause mortality	SHARP 2011 Multi[24]	5 y (4 y)	Simvastatin 20 mg/d + ezetimibe 10 mg/d	Placebo	4650 (4650)	4620 (4620)	GFR 27 (27) mL/min/ 1.73 m^2	23 (23)	4.88 (4.90)	2.77 (2.78)	1.12 (1.11)	2.31 (2.34)	1142 (25%) [1115 (24%)]	RR 1.02 (0.94, 1.11)	NS	Good
Death from any cancer													150 (3%) [128 (3%)]	RR 1.17 (0.92, 1.48)	nd	Good
Death from renal causes													164 (4%) [173 (4%)]	RR 0.95 (0.76, 1.17)	nd	Good
Death from respiratory causes													124 (3%) [100 (2%)]	RR 1.24 (0.95,1.61)	nd	Good
Death from GI causes													72 (2%) [70 (2%)]	RR 1.03 (0.74, 1.42)	nd	Good
Death from other medical causes													124 (3%) [119 (3%)]	RR 1.04 (0.81, 1.34)	nd	Good
Death from trauma or fracture													34 (1%) [22 (1%)]	RR 1.53 (0.91, 2.59)	nd	Good
Any nonvascular death													668 (14%) [612 (13%)]	RR 1.09 (0.98, 1.21)	NS	Good
Sudden death													50 (1%) [55 (1%)]	RR 0.91 (0.62, 1.33)	nd	Good
Death for reasons unclear													63 (1%) [60 (1%)]	RR 1.05 (0.74, 1.49)	nd	Good
Death from unknown causes													113 (2%) [115 (3%)]	RR 0.98 (0.76, 1.27)	NS	Good
CV mortality																
CHD death	SHARP 2011 Multi[24]	5 y (4 y)	Simvastatin 20 mg/d +	Placebo	4650 (4650)	4620 (4620)	GFR 27 (27)	23 (23)	4.88 (4.90)	2.77 (2.78)	1.12 (1.11)	2.31 (2.34)	91 (2%) [90 (2%)]	RR 1.01 (0.75, 1.35)	NS	Good

Cardiac death other than CHD			ezetimibe 10 mg/d				mL/min/ 1.73 m^2						162 (4%) [182 (4%)]	RR 0.89 (0.72, 1.09)	nd	Good
All cardiac deaths													253 (5%) [272 (6%)]	RR 0.93 (0.78, 1.10)	NS	Good
Death from ischemic stroke													30 (1%) [41 (1%)]	RR 0.73 (0.46, 1.16)	nd	Good
Death from haemorrhagic stroke													27 (1%) [23 (1%)]	RR 1.16 (0.67, 2.03)	nd	Good
Death from unspecified stroke													11 (0.2%) [14 (0.3%)]	RR 0.78 (0.36, 1.71)	nd	Good
Death from any type of stroke													68 (2%) [78 (2%)]	RR 0.87 (0.63, 1.20)	NS	Good
Any other vascular death													40 (1%) [38 (1%)]	RR 1.05 (0.67, 1.64)	NS	Good
Any vascular death (total)													361 (8%) [388 (8%)]	RR 0.93 (0.80, 1.07)	NS	Good
CV event																
Any major coronary event													213 (5%) [230 (5%)]	RR 0.92 (0.76, 1.11)	NS	Good
Non-fatal MI													134 (3%) [159 (3%)]	RR 0.84 (0.66, 1.05)	NS	Good
Any non-haemorrhagic stroke													131 (3%) [174 (4%)]	RR 0.75 (0.60, 0.94)	0.01	Good
Ischemic stroke													114 (3%) [157 (3%)]	RR 0.72 (0.57, 0.92)	0.0073	Good
Unknown stroke	SHARP 2011 Multi[24]	5 y (4 y)	Simvastatin 20 mg/d + ezetimibe 10 mg/d	Placebo	4650 (4650)	4620 (4620)	GFR 27 (27) mL/min/ 1.73 m^2	23 (23)	4.88 (4.90)	2.77 (2.78)	1.12 (1.11)	2.31 (2.34)	18 (0.4%) [19 (0.4%)]	RR 0.94 (0.49, 1.79)	NS	Good
Any stroke													171 (4%) [210 (5%)]	RR 0.81 (0.66, 0.99)	0.04	Good
Haemorrhagic stroke													45 (1%) [37 (1%)]	RR 1.21 (0.78, 1.86)	NS	Good
Any revascularization procedure													284 (6%) [352 (8%)]	RR 0.79 (0.68, 0.93)	0.0036	Good
Coronary revascularization													149 (3%) [203 (4%)]	RR 0.73 (0.59, 0.90)	0.0027	Good

Non-coronary revascularization													154 (3%) [169 (4%)]	RR 0.90 (0.73, 1.12)	NS	Good
First major atherosclerotic events[89]													596 (11%) [619 (13%)]	RR 0.83 (0.74, 0.94)	0.0021	Good
Major vascular events[90]													701 (15%) [814 (18%)]	RR 0.85 (0.77, 0.94)	0.0012	Good
PCI													106 (2%) [148 (3%)]	RR 0.71 (0.56, 0.91)	0.0063	Good
CABG													50 (1%) [66 (1%)]	RR 0.75 (0.52, 1.09)	0.13	Good
Major atherosclerotic events in CKD 1-2					44 (44)	44 (44)	GFR ≥ 60 mL/min/ 1.73 m^2						3 (7%) [3 (7%)]	RR 0.84 (0.17, 4.18)	nd	Good
Major atherosclerotic events in CKD 3					1100 (1100)	1055 (1055)	GFR ≥30-<60 mL/min/ 1.73 m^2						87 (8%) [110 (10%)]	RR 0.75 (0.57, 1.00)	nd	Good
Major atherosclerotic events in CKD 4					1246 (1246)	1319 (1319)	GFR ≥15-<30 mL/min/ 1.73 m^2						127 (10%) [168 (13%)]	RR 0.78 (0.62, 0.98)	nd	Good
Major atherosclerotic events in CKD 5					614 (614)	607 (607)	GFR <15 mL/min/ 1.73 m^2	nd	nd	nd	nd	nd	67 (11%) [81 (13%)]	RR 0.82 (0.59, 1.13)	nd	Good
Major atherosclerotic events in CKD 5HD					1275 (1275)	1252 (1252)							194 (15%) [199 (16%)]	RR 0.95 (0.78, 1.15)	nd	Good
Major atherosclerotic events in CKD 5PD					258 (258)	238 (238)	nd						36 (14%) [47 (20%)]	RR 0.70 (0.46, 1.08)	nd	Good
Major atherosclerotic events in CKD 5D					1533 (1533)	1490 (1490)							230 (15%) [246 (17%)]	RR 0.90 (0.75, 1.08)	nd	Good

[89] Non-fatal MI or coronary death, nonhaemorrhagic stroke, or arterial revascularization

[90] Major atherosclerotic events plus non-coronary cardiac deaths and haemorrhagic strokes)

Kidney function																
ESRD defined as dialysis or transplantation	SHARP 2011 Multi[24]	5 y (4 y)	Simvastatin 20 mg/d + ezetimibe 10 mg/d	Placebo	4650 (4650)	4620 (4620)	GFR 27 (27) mL/min/ 1.73 m^2	23 (23)	4.88 (4.90)	2.77 (2.78)	1.12 (1.11)	2.31 (2.34)	1057 (34%) [1084 (35%)]	RR 0.97 (0.89, 1.05)	NS	Good
ESRD or doubling of SCr													1190 (38%) [1257 (40%)]	RR 0.93 (0.86, 1.01)	NS (0.09)	Good

Supplemental Table 18: Summary table of RCT examining the effect of dose of atorvastatin in CKD patients with DM [categorical outcomes]

Outcome	Author, Year Country Ref #	Duration Outcome measure-ment (Treatment)	Description: Intervention	Description: Control	No. Analyzed (Enrolled): Intervention	No. Analyzed (Enrolled): Control	Baseline: eGFR	Baseline: DM (%)	Baseline: TC (mg/dL)	Baseline: LDL (mg/dL)	Baseline: HDL (mg/dL)	Baseline: Tg (mg/dL)	Results*: Events No. (%) Intervention [Control]	Results*: RR/HR[92] (95% CI)	P value[91]	Quality
Mortality																
All-cause mortality	TNT Shepherd 2008 Multi[25]	Median 5 y (5 y)	80-mg Atorvastatin	10-mg Atorvastatin	273 (273)	273 (273)	51.5 (50.7) mL/min/ 1.73 m[2]	100 (100)	176.1 (178.0)	95.5 (97.0)	44.9 (45.2)	181.6 (180.2)	33 (12%) [32 (12%)]	RR 1.03 (0.65, 1.63)	NS	Good
All-cause mortality	TNT Shepherd 2008 Multi[26]	Median 5 y (5 y)	80-mg Atorvastatin	10-mg Atorvastatin	1602 (1602)	1505 (1505)	53.0 (52.8) mL/min/ 1.73 m[2]	17 (18)	175.9 (175.9)	96.3 (96.5)	48.0 (47.6)	159.2 (159.8)	112 (7%) [113 (8%)]	HR 0.95 (0.70, 1.2)[93]	NS	Good
CV events																
Major CV event (primary)[94]	TNT Shepherd 2008 Multi[25]	Median 5 y (5 y)	80-mg Atorvastatin	10-mg Atorvastatin	273 (273)	273 (273)	51.5 (50.7) mL/min/ 1.73 m[2]	100 (100)	176.1 (178.0)	95.5 (97.0)	44.9 (45.2)	181.6 (180.2)	38 [14%) [57 (21%)]	HR 0.65 (0.43, 0.98)	0.04	Good
Any CV event													120 (44%) [140 (51%)]	RR 0.86 (0.72, 1.02)	NS (0.08)	Good
Major coronary event[95]													28 (10%) [43 (16%)]	RR 0.65 (0.42, 1.02)	NS (0.06)	Good
Any coronary event													80 (29%) [97 (36%)]	RR 0.82 (0.65, 1.05)	NS	Good
Cerebrovascular event													24 (9%) [36 (13%)]	RR 0.67 (0.41, 1.09)	NS	Good
Stroke													13 (5%) [20 (7%)]	RR 0.65 (0.33, 1.28)	NS	Good

[91] Calculated by ERT for all outcome other than primary outcome of major CV event
[92] Calculated by ERT for all outcome other than primary outcome of major CV event
[93] Estimated from figure
[94] Death from CHD, nonfatal non-procedure-related MI, resuscitation after cardiac arrest, or fatal or nonfatal stroke.
[95] Death from CHD, nonfatal non-procedure-related MI, or resuscitation after cardiac arrest.

Outcome	Author, Year Country Ref #	Duration Outcome measure-ment (Treatment)	Description		No. Analyzed (Enrolled)		Baseline						Results*		P value[91]	Quality
			Intervention	Control	Intervention	Control	eGFR	DM (%)	TC	LDL	HDL	Tg	Events No. (%) Intervention [Control]	RR/HR[92] (95% CI)		
									mg/dL							
CHF with hospitalization													25 (9%) [34 (13%)]	RR 0.74 (0.45, 1.20)	NS	Good
Peripheral artery disease													35 (13%) [30 (11%)]	RR 1.17 (0.74, 1.84)	NS	Good
Major CV event (primary)													149 (9%) [202 (13%)]	HR 0.68 (0.55, 0.84)	0.0003	Good
Any CV event													489 (31%) [574 (38%)]	HR 0.76 (0.67, 0.86)	nd	Good
Major coronary event													110 (7%) [157 (10%)	HR 0.65 (0.51, 0.83)	nd	Good
Any coronary event	TNT Shepherd 2008 Multi[26]				1602 (1602)	1505 (1505)	53.0 (52.8) mL/min/ 1.73 m^2	17 (18)	175.9 (175.9)	96.3 (96.5)	48.0 (47.6)	159.2 (159.8)	356 (22%) [431 (29%)]	HR 0.75 (0.65, 0.86)	nd	Good
Cerebrovascular event													74 (5%) [104 (7%)]	HR 0.66 (0.49, 0.89)	nd	Good
CHF with hospitalization													49 (3%) [84 (6%)]	HR 0.54 (0.38, 0.77)	nd	Good
Peripheral artery disease													121 (8%) [112 (7%)]	HR 1.0 (0.8, 1.4)[96]	nd	Good

96 Estimated from figure

Supplemental Table 19: Drug interactions

		Simvastatin	Lovastatin	Atorvastatin	Fluvastatin	Pravastatin	Rosuvastatin	Pitavastatin	Reference #
Inhibitors (increase statin AUC)	Potent CYP3A4 inhibitors (eg, itraconazole)	5- to 20-fold increase	5- to 20-fold increase	2- to 4-fold increase	<1.5-fold	<1.5-fold	<1.5-fold	Unchanged	[27-32]
	Moderate CYP3A4 inhibitors (eg, clarithromycin and diltiazem)	4- to 12-fold increase	2- to 10-fold increase	1- to 5-fold increase	Unchanged	<2-fold	Unchanged	Unchanged	[33, 34]
	Cyclosporine (an OATP1B1 and CYP3A$ inhibitor)	6- to 10-fold incrase	5- to 20-fold increase	6- to 15-fold increase	2- to 4-fold increase	5- to 10-fold increase	5- to 10-fold increase	5-fold increase	[32, 35-37]
	Gemfibrozil (a CYP2C8 and OATP1B1 inhibitor)	2- to 3-fold increase	2- to 3-fold increase	<1.5-fold increase	unchanged	2-fold increase	2-fold increase	≤1.5-fold increase	[38-44]
Inducers (decrease statin AUC)	Potent inducers of CYP3A4 (eg, rifampicin and carbamazepine)	70 to 95% decrease	70 to 95% decrease	60 to 90% decrease	50% decrease	30% decrease	Unchanged	Unknown	[39]

Supplemental Table 20: From Vaquero et al. 2010[45]

Effect of grapefruit juice (GFJ) on statin pharmacokinetics and recommendations				
Reference	*Statin*	*Treatments*	*Effects*	*Recommendations*
Lilja et al.[16]	Simvastatin	200 ml GFJ 3 times/day. On the third day 40 mg of Simvastatin with the first GFJ intake	↑ C_{max} and ↑ AUC of statins (lactone and acid forms)	Avoid concomitant use of GFJ and Simvastatin, or reduce Simvastatin dose
Lilja et al.[17] Fukazawa et al.[19]	Atorvastatin Pravastatin	200 or 250 ml GFJ 3 times/day. On the third day Atorvastatin (10 or 20 mg) or Pravastatin (10 mg) with the first GFJ intake	↑ C_{max} and ↑ AUC (0-48 h) of Atorvastatin No significant changes in Pravastatin pharmacokinetics	Use Pravastatin which presents little or not interaction with GFJ, or reduce Atorvastatin dose
Ando et al.[20]	Atorvastatin Pitavastatin	250 ml GFJ 3 times/day for 4 days. On each day 20 mg of Atorvastatin or 4 mg Pitavastatin with the first GFJ intake	↑ AUC of Pitavastatin acid was only 13% while that of Atorvastatin was 83%	Use Pitavastatin
Rogers et al.[23]	Lovastatin	250 ml GFJ once daily for 3 days 12 h apart from Lovastatin	No significant changes in Lovastatin pharmacokinetics. High variability in the individual response	Lovastatin taken 12 h apart form GFJpresents a modest interaction
Lilja et al.[18]	Simvastatin	200 ml GFJ once daily for 3 days. On the third day 40 mg of Simvastatin with the GFJ intake	↑ C_{max} and ↑ AUC (0-24 h)	Even one glass of GFJ increases Simvastatin plasma concentrations
Lilja et al.[24]	Simvastatin	200 ml double-strength GFJ 3 times/day for 3 days with 40 mg Simvastatin, or 1, 3, and 7 days after ingestion of the high-dose GFJ.	↑ C_{max} and ↑ AUC was lower when Simvastatin was taken after 24 h of GFJ intake, even less after 3 days, and not significant after 7 days	The interaction of large amounts of GFJ with Simvastatin disappears after 3-7 days

Supplemental Table 21: Patients on statin + fibrate therapy reporting any adverse event

Study Year Country Ref #	Population	Duration Outcome measurement (Treatment)	Description		No analyzed (Enrolled)		Baseline GFR or SCr	% with DM	Results			P value
			Intervention	Control	Intervention	Control			Definition	No. (%) Intervention [Control]	RR[97] (95% CI)	
Wiklund 1993 Sweden & Finland[46]	Adults with hypercholesterolemia	12 wk (12 wk)	Pravastatin + gemfibrozil	Placebo	75 (75)	73 (73)	nd	nd	Any AE	31 (41%) [21 (29%)]	1.44 (0.92, 2.25)	nd
				Pravastatin		70 (70)				31 (41%) [16 (23%)]	1.81 (1.09, 3.00)	nd
				Gemfibrozil		72 (72)				31 (41%) [25 (35%)]	1.19 (0.79, 1.80)	nd
Roth 2010 US[47]	Adults with hypercholesterolemia and hypertriglyceridemia	12 wk (8 wk)	5 mg rosuvastatin + fenofibric acid	Simvastatin[98]	118 (118)	119 (119)	nd	22 (16)	Treatment emergent AE	53 (45%) [58 (49%)]	0.92 (0.70, 1.21)	nd
			10 mg rosuvastatin + fenofibric acid		119 (119)			19 (16)		45 (38%) [58 (49%)]	0.78 (0.58, 1.04)	nd
			20 mg rosuvastatin + fenofibric acid		118 (118)			23 (16)		71 (60%) [58 (49%)]	1.23 (0.98, 1.56)	nd
Jones 2010 US & Canada[48] [Subgroup analysis of 3 RCTs]	Adults with mixed dyslipidemia	2 y (2 y)	Fenofibric acid + low-dose rosuvastatin, simvastatin, or atorvastatin	Fenofibric acid	106 (106)	105 (105)	nd	100 (100)	Any treatment-emergent AE	82 (77%) [70 (67%)]	1.16 (0.98, 1.38)	nd
				Low-dose statin only		105 (105)				82 (77%) [57 (54%)]	1.43 (1.16, 1.75)	nd
				Moderate-dose statin only		107 (107)				82 (77%) [72 (67%)]	1.15 (0.97, 1.36)	nd
			Fenofibric acid + moderate-dose rosuvastatin, simvastatin, or atorvastatin	Fenofibric acid	110 (110)	105 (105)				83 (76%) [70 (67%)]	1.13 (0.95, 1.34)	nd
				Low-dose statin only		105 (105)				83 (76%) [57 (54%)]	1.39 (1.13, 1.71)	nd

[97] Calculated by ERT
[98] FDA issued a Drug Safety Communication on June 8, 2011, about limiting use of the highest approved dose, 80 mg, because of increased risk of muscle damage. (See http://www.fda.gov/Drugs/DrugSafety/ucm256581.htm).

Study Year Country Ref #	Population	Duration Outcome measurement (Treatment)	Description		No analyzed (Enrolled)		Baseline GFR or SCr	% with DM	Results			P value
			Intervention	Control	Intervention	Control			Definition	No. (%) Intervention [Control]	RR[97] (95% CI)	
				Moderate dose statin only		107 (107)				83 (76%) [72 (67%)]	1.12 (0.95, 1.33)	nd
Jones 2010 US[49]	Adults with mixed dyslipidemia	16 wk (12 wk)	Fenofibric acid + atorvastatin and ezetimibe	Placebo + atorvastatin and ezetimibe	272 (272)	270 (271)	nd	20 (23)	Any AE	138 (51%) [137 (51%)]	1.00 (0.85, 1.18)	nd
Farnier 2010 EU[50]	Adults with mixed dyslipidemia	12 wk (12 wk)	Fenofibrate + pravastatin	Pravastatin	123 (123)	125 (125)	S_{Cr} 84.2 (8.5) µmol/L	26 (29)	Any AE	45 (37%) [40 (32%)]	1.14 (0.81, 1.62)	nd
Farnier 2007 Multi[51]	Adults with mixed hyperlipidemia	12 wk (12 wk)	Ezetimibe + simvastatin and fenofibrate	Ezetimibe + simvastatin	183 (183)	184 (184)	Median S_{Cr} 1.1 (1.1) mg/dL	12 (6)	One or more AEs	72 (39%) [65 (35%)]	1.11 (0.85,1.45)	nd
				Fenofibrate		184 (184)	Median S_{Cr} 1.1 (1.0) mg/dL	12 (9)		72 (39%) [87 (47%)]	0.83 (0.66, 1.05)	nd
				Placebo		60 (60)	Median S_{Cr} 1.1 (1.0) mg/dL	12 (10)		72 (39%) [18 (30%)]	5.97 (3.69, 9.66)	nd
Davidson 2009 US[52]	Adults with dyslipidemia	12 wk (12 wk)	Atorvastatin + fenofibrate	Atorvastatin	73 (73)	74 (74)	S_{Cr} 1.0 (0.9) mg/dL	nd	Treatment-emergent AE	43 (59%) [49 (66%)]	0.89 (0.69, 1.14)	nd

Study Year Country Ref #	Population	Duration Outcome measurement (Treatment)	Description		No analyzed (Enrolled)		Baseline GFR or SCr	% with DM	Results			P value
			Intervention	Control	Intervention	Control			Definition	No. (%) Intervention [Control]	RR[97] (95% CI)	
				Fenofibrate		73 (73)	SCr 1.0 (1.0) mg/dL			43 (59%) [48 (66%)]	0.90 (0.70, 1.15)	nd

Supplemental Table 22: Patients receiving statin + fibrate therapy reporting other individual adverse events

Study Year Country Ref #	Population	Duration Outcome measurement (Treatment)	Description		No analyzed (Enrolled)		Baseline GFR or SCr	% with DM	Results			P value
			Intervention	Control	Intervention	Control			Definition	No. (%) Intervention [Control]	RR (95% CI)[99]	
Wiklund 1993 Sweden & Finland[46]	Adults with hypercholesterolemia	12 wk (12 wk)	Pravastatin + gemfibrozil	Placebo	75 (75)	73 (73)	nd	nd	Abdominal pain	5 (7%) [0 (0%)]	--	nd
				Pravastatin		70 (70)				5 (7%) [0 (0%)]	--	nd
				Gemfibrozil		72 (72)				5 (7%) [7 (10%)]	0.69 (0.23, 2.06)	nd
			Pravastatin + gemfibrozil	Placebo	75 (75)	73 (73)			Diarrhea	4 (5%) [2 (3%)]	1.95 (0.37, 10.31)	nd
				Pravastatin		70 (70)				4 (5%) [0 (0%)]	--	nd
				Gemfibrozil		72 (72)				4 (5%) [2 (3%)]	1.92 (0.36, 10.16)	nd
			Pravastatin + gemfibrozil	Placebo	75 (75)	73 (73)			Dizziness	2 (3%) [0 (0%)]	--	nd
				Pravastatin		70 (70)				2 (3%) [1 (1%)]	1.87 (0.17, 20.13)	nd
				Gemfibrozil		72 (72)				2 (3%) [2 (3%)]	0.96 (0.14, 6.63)	nd
			Pravastatin + gemfibrozil	Placebo	75 (75)	73 (73)			Dyspepsia/heartburn	0 (0%) [0 (0%)]	--	nd
				Pravastatin		70 (70)				0 (0%) [1 (1%)]	--	nd
				Gemfibrozil		72 (72)				0 (0%) [3 (4%)]	--	nd
			Pravastatin + gemfibrozil	Placebo	75 (75)	73 (73)			Epigastric pain	2 (3%) [2 (3%)]	0.97 (0.14, 6.73)	nd
				Pravastatin		70 (70)				2 (3%) [1 (1%)]	1.87 (0.17, 20.13)	nd
				Gemfibrozil		72 (72)				2 (3%) [0 (0%)]	--	nd
			Pravastatin + gemfibrozil	Placebo	75 (75)	73 (73)			Fatigue	0 (0%) [1 (1%)]	--	nd
				Pravastatin		70 (70)				0 (0%) [1 (1%)]	--	nd
				Gemfibrozil		72 (72)				0 (0%) [2 (3%)]	--	nd

[99] All RRs and CIs were calculated by ERT.

Study Year Country Ref #	Population	Duration Outcome measurement (Treatment)	Description		No analyzed (Enrolled)		Baseline GFR or SCr	% with DM	Results			P value
			Intervention	Control	Intervention	Control			Definition	No. (%) Intervention [Control]	RR (95% CI)[99]	
			Pravastatin + gemfibrozil	Placebo	75 (75)	73 (73)			Fever	1 (1%) [3 (3%)]	0.32 (0.03, 3.05)	nd
				Pravastatin		70 (70)				1 (1%) [1 (1%)]	0.93 (0.06, 14.64)	nd
				Gemfibrozil		72 (72)				1 (1%) [0 (0%)]	--	nd
			Pravastatin + gemfibrozil	Placebo	75 (75)	73 (73)			Gastritis	1 (1%) [0 (0%)]	--	nd
				Pravastatin		70 (70)				1 (1%) [0 (0%)]	--	nd
				Gemfibrozil		72 (72)				1 (1%) [6 (8%)]	0.16 (0.02, 1.30)	nd
			Pravastatin + gemfibrozil	Placebo	75 (75)	73 (73)			Headache	1 (1%) [3 (4%)]	0.32 (0.03, 3.05)	nd
				Pravastatin		70 (70)				1 (1%) [0 (0%)]	--	nd
				Gemfibrozil		72 (72)				1 (1%) [0 (0%)]	--	nd
			Pravastatin + gemfibrozil	Placebo	75 (75)	73 (73)			Malaise	2 (3%) [0 (0%)]	--	nd
				Pravastatin		70 (70)				2 (3%) [0 (0%)]	--	nd
				Gemfibrozil		72 (72)				2 (3%) [2 (3%)]	0.96 (0.14, 6.63)	nd
			Pravastatin + gemfibrozil	Placebo	75 (75)	73 (73)			Musculoskeletal pain	7 (9%) [3 (4%)]	2.27 (0.61, 8.45)	nd
				Pravastatin		70 (70)				7 (9%) [1 (1%)]	6.53 (0.82, 51.77)	nd
				Gemfibrozil		72 (72)				7 (9%) [3 (4%)]	2.24 (0.60, 8.33)	nd
			Pravastatin + gemfibrozil	Placebo	75 (75)	73 (73)			Upper respiratory infection	6 (8%) [5 (7%)]	1.17 (0.37, 3.66)	nd
				Pravastatin		70 (70)				6 (8%) [2 (3%)]	2.80 (0.58, 13.42)	nd
				Gemfibrozil		72 (72)				6 (8%) [3 (4%)]	1.92 (0.50, 7.39)	nd
Farnier 2010 EU[50]	Adults with mixed dyslipidemia	12 wk (12 wk)	Fenofibrate + pravastatin	Pravastatin	123 (123)	125 (125)	S_{Cr} 84.2 (85.5) μmol/L	26 (29)	GI AEs	5 (4%) [4 (3%)]	1.27 (0.35, 4.62)	nd
									Musculoskeletal AE	5 (4%) [4 (3%)]	1.27 (0.35, 4.62)	nd

Study Year Country Ref #	Population	Duration Outcome measurement (Treatment)	Description		No analyzed (Enrolled)		Baseline GFR or SCr	% with DM	Results			P value
			Intervention	Control	Intervention	Control			Definition	No. (%) Intervention [Control]	RR (95% CI)[99]	
									Myalgia	8 (7%) [8 (6%)]	1.02 (0.39, 2.62)	nd
									Nervous system AE	3 (2%) [2 (2%)]	1.52 (0.26, 8.97)	nd
Farnier 2007 Multi[51]	Adults with mixed hyperlipidemia	12 wk (12 wk)	Ezetimibe + simvastatin and fenofibrate	Ezetimibe +, simvastatin	183 (183)	184 (184)	Median S_{Cr} 1.1 (1.1) mg/dL	12 (6)	Drug-related GI AE	7% [9%]	--	nd
				Fenofibrate		184 (184)	Median S_{Cr} 1.1 (1.0) mg/dL	12 (9)		7% [10%]	--	nd
				Placebo		60 (60)	Median S_{Cr} 1.1 (1.0) mg/dL	12 (6)		7% [7%]	--	nd
			Ezetimibe + simvastatin and fenofibrate	Ezetimibe +, simvastatin	183 (183)	184 (184)	Median S_{Cr} 1.1 (1.1) mg/dL	12 (6)	Myalgia	8 (4%) [7 (4%)]	1.15 (0.43, 3.10)	nd
				Fenofibrate		184 (184)	Median S_{Cr} 1.1 (1.0) mg/dL	12 (9)		8 (4%) [6 (3%)]	1.34 (0.47, 3.79)	nd
				Placebo		60 (60)	Median S_{Cr} 1.1 (1.0) mg/dL	12 (6)		8 (4%) [0 (0%)]	--	nd
			Ezetimibe + simvastatin and fenofibrate	Ezetimibe +, simvastatin	183 (183)	184 (184)	Median S_{Cr} 1.1 (1.1) mg/dL	12 (6)	Rash	2% [1%]	--	nd
				Fenofibrate		184 (184)	Median S_{Cr} 1.1 (1.0) mg/dL	12 (9)		2% [5%]	--	nd

Study Year Country Ref #	Population	Duration Outcome measurement (Treatment)	Description		No analyzed (Enrolled)		Baseline GFR or SCr	% with DM	Results			P value
			Intervention	Control	Intervention	Control			Definition	No. (%) Intervention [Control]	RR (95% CI)[99]	
				Placebo		60 (60)	Median S_{Cr} 1.1 (1.0) mg/dL	12 (6)		2% [7%]	--	nd
Roth 2010 US[47]	Adults with hypercholesterolemia and hypertriglyceridemia	12 wk (8 wk)	5 mg rosuvastatin + fenofibric acid	Simvastatin	118 (118)	119 (119)	nd	22 (16)	Muscle, hepatic, and renal AEs	3 (3%) [0 (0%)]	--	nd
			10mg rosuvastatin + fenofibric acid		119 (119)			19 (16)		3 (3%)) [0 (0%)]	--	nd
			20 mg rosuvastatin + fenofibric acid		118 (118)			23 (16)		5 (4%) [0 (0%)]	--	nd
Davidson 2009 US[52]	Adults with dyslipidemia	12 wk (12 wk)	Atorvastatin + fenofibrate	Atorvastatin	73 (73)	74 (74)	S_{Cr} 1.0 (0.9) mg/dL	nd	Muscle-associated AEs except myalgia	2 (3%) [5 (7%)]	0.41 (0.08, 2.02)	nd
				Fenofibrate		73 (73)	S_{Cr} 1.0 (0.9) mg/dL			2 (3%) [3 (4%)]	0.67 (0.11, 3.87)	nd
			Atorvastatin + fenofibrate	Atorvastatin	73 (73)	74 (74)	S_{Cr} 1.0 (0.9) mg/dL	nd	Myalgia	0 (0%) [2 (3%)]	--	nd
				Fenofibrate		73 (73)	S_{Cr} 1.0 (0.9) mg/dL			0 (0%) [2 (3%)]	--	nd
Yang 2009 New Zealand[53] [Post hoc analysis of 3 RCTs]	Adults with mixed dyslipidemia	Median 12 wk (Median 12 wk)	Fenofibric acid+ low-dose atorvastatin, rosuvastatin, or simvastatin	Atorvastatin, rosuvastatin, or simvastatin	1895 (1911)		nd	19–26	Myalgia[100]	2–5% [3–5%]	--	nd
				Fenofibric acid						2–5% [3–7%]	--	nd

[100] Explicit report of "no myalgia": Roth et al 2010[47]. Roth EM, McKenney JM, Kelly MT, *et al.* Efficacy and safety of rosuvastatin and fenofibric acid combination therapy versus simvastatin monotherapy in patients with hypercholesterolemia and hypertriglyceridemia: a randomized, double-blind study. *Am J Cardiovasc Drugs* 2010; **10:** 175-186.; Feher et al 1995[54]. Feher MD, Foxton J, Banks D, *et al.* Long-term safety of statin-fibrate combination treatment in the management of hypercholesterolaemia in patients with coronary artery disease. *Br Heart J* 1995; **74:** 14-17.

Study Year Country Ref #	Population	Duration Outcome measurement (Treatment)	Description		No analyzed (Enrolled)		Baseline GFR or SCr	% with DM	Results			P value
			Intervention	Control	Intervention	Control			Definition	No. (%) Intervention [Control]	RR (95% CI)[99]	
Jones 2010 US & Canada[48] [Subgroup analysis of 3 RCTs]	Adults with mixed dyslipidemia	2 y (2 y)	Fenofibric acid+ low dose rosuvastatin, simvastatin, or atorvastatin	Fenofibric acid	106 (106)	105 (105)	nd	100 (100)	Myalgia	2 (2%) [1 (1%)]	1.98 (0.18, 21.52)	nd
				Low dose statin		105 (105)				2 (2%) [6 (6%)]	0.33 (0.07, 1.60)	nd
				Moderate dose statin		107 (107)				2 (2%) [2 (2%)]	1.01 (0.14, 7.04)	nd
			Fenofibric acid+ moderate dose rosuvastatin, simvastatin, or atorvastatin	Fenofibric acid	110 (110)	105 (105)				3 (2%) [1 (1%)]	2.86 (0.30, 27.10)	nd
				Low dose statin		105 (105)				3 (2%) [6 (6%)]	0.95 (0.20, 4.62)	nd
				Moderate dose statin		107 (107)				3 (2%) [2 (2%)]	1.46 (0.25, 8.56)	nd
Jones 2010 US[49]	Adults with mixed dyslipidemia	16 wk (12 wk)	Fenofibric acid + atorvastatin and ezetimibe	Placebo + atorvastatin and ezetimibe	272 (272)	270 (271)	nd	20 (23)	Myalgia	3% [4%]	--	NS
Athyros 2001 Greece[55]	Patients with familial combined hyperlipidemia, 48% with and 52% without CAD at entry	1 y	Simvastatin + gemfibrozil	Atorvastatin	135 (135)	134 (134)	nd	100 (100)	Myalgia without CK elevation	1 (1%) [0 (0%)]	--	nd
			Pravastatin + gemfibrozil		136 (136)					1 (1%) [0 (0%)]	--	nd

Supplemental Table 23: Patients on statin + fibrate therapy reporting treatment related adverse events

Study Year Country Ref #	Population	Duration Outcome measurement (Treatment)	Description		No analyzed (Enrolled)		Baseline GFR or SCr	% with DM	Results			P value
			Intervention	Control	Intervention	Control			Definition	No. (%) Intervention [Control]	RR (95% CI)[101]	
Yang 2009 New Zealand[53] [Post hoc analysis of 3 RCTs]	Adults with mixed dyslipidemia	Median 12 wk (Median 12 wk)	Fenofibric acid+ low- or moderate-dose atorvastatin, rosuvastatin, or simvastatin	Atorvastatin, rosuvastatin, or simvastatin	1895 (1911)		nd	19–26	Treatment-related AEs	23–24% [16–24%]	--	NS
				Fenofibric acid						23–24% [13–33%]	--	NS
			Fenofibric acid + atorvastatin	Atorvastatin						20% [6%]	--	0.003
			Fenofibric acid + rosuvastatin	Rosuvastatin						27% [17%]	--	0.006
			Fenofibric acid +low-dose atorvastatin, rosuvastatin, or simvastatin	Low-dose atorvastatin, rosuvastatin, or simvastatin						25% [14%]	--	<0.05
Roth 2010 US[47]	Adults with hypercholesterolemia and hypertriglyceridemia	12 wk (8 wk)	5 mg rosuvastatin + fenofibric acid	Simvastatin	118 (118)	119 (119)	nd	22 (16)	Treatment-related AE	16 (14%) [20 (17%)]	0.81 (0.44, 1.48)	nd
			10 mg rosuvastatin + fenofibric acid		119 (119)			19 (16)		17 (14%) [20 (17%)]	0.85 (0.47, 1.54)	nd

[101] All RRs and CIs were calculated by ERT.

Study Year Country Ref #	Population	Duration Outcome measurement (Treatment)	Description		No analyzed (Enrolled)		Baseline GFR or SCr	% with DM	Results			P value
			Intervention	Control	Intervention	Control			Definition	No. (%) Intervention [Control]	RR (95% CI)[101]	
			20 mg rosuvastatin + fenofibric acid		118 (118)			23 (16)		18 (15%) [20 (17%)]	0.91 (0.51, 1.63)	nd
Jones 2010 US[49]	Adults with mixed dyslipidemia	16 wk (12 wk)	Fenofibric acid + atorvastatin and ezetimibe	Placebo + atorvastatin and ezetimibe	272 (272)	270 (271)	nd	20 (23)	Treatment-related AE	52 (19%) [45 (17%)]	1.15 (0.80, 1.65)	nd
Farnier 2010 EU[50]	Adults with mixed dyslipidemia	12 wk (12 wk)	Fenofibrate + pravastatin	Pravastatin	123 (123)	125 (125)	S_{Cr} 84.2 (85.5) µmol/L	26 (29)	Drug-related AE	13 (11%) [12 (10%)]	1.10 (0.52, 2.32)	nd
									Serious drug-related AE	1 (1%) [0 (0%)]	--	--
Farnier 2007 Multi[51]	Adults with mixed hyperlipidemia	12 wk (12 wk)	Ezetimibe + simvastatin and fenofibrate	Ezetimibe + simvastatin	183 (183)	184 (184)	Median S_{Cr} 1.1 (1.1) mg/dL	12 (6)	Drug-related AEs	16 (9%) [13 (7%)]	1.24 (0.61, 2.50)	nd
				Fenofibrate		184 (184)	Median S_{Cr} 1.1 (1.0) mg/dL	12 (9)		16 (9%) [23 (13%)]	0.70 (0.38, 1.28)	nd
				Placebo		60 (60)	Median S_{Cr} 1.1 (1.0) mg/dL	12 (10)		16 (9%) [4 (7%)]		nd
Davidson 2009 US[52]	Adults with dyslipidemia	12 wk (12 wk)	Atorvastatin + fenofibrate	Atorvastatin	73 (73)	74 (74)	S_{Cr} 1.0 (0.9) mg/dL	nd	Treatment emergent AEs related	12 (16%) [17 (23%)]	0.72 (0.37, 1.39)	nd

Study Year Country Ref #	Population	Duration Outcome measurement (Treatment)	Description		No analyzed (Enrolled)		Baseline GFR or SCr	% with DM	Results			P value
			Intervention	Control	Intervention	Control			Definition	No. (%) Intervention [Control]	RR (95% CI)[101]	
				Fenofibrate		73 (73)	S_{Cr} 1.0 (1.0) mg/dL		to the study drug	12 (16%) [19 (26%)]	0.63 (0.33, 1.20)	nd

Supplemental Table 24: Patients on statin + fibrate therapy discontinuing due to adverse events

Study Year Country Ref #	Population	Duration Outcome measurement (Treatment)	Description		No analyzed (Enrolled)		Baseline GFR or SCr	% with DM	Results			P value
			Intervention	Control	Intervention	Control			Definition	No. (%) Intervention [Control]	RR (95% CI)[102]	
Yang 2009 New Zealand[53] [Post hoc analysis of 3 RCTs]	Adults with mixed dyslipidemia	Median 12 wk (Median 12 wk)	Fenofibric acid+ low-dose atorvastatin, rosuvastatin, or simvastatin	Atorvastatin, rosuvastatin, or simvastatin	1895 (1911)		nd	19–26	AEs leading to study discontinuation	9% [4%]	--	≤0.05
			Fenofibric acid + atorvastatin	Atorvastatin						11% [3%]	--	≤0.05
			Fenofibric acid + 10 mg/d rosuvastatin	10 mg/d Rosuvastatin						10% [4%]	--	≤0.05
			Fenofibric acid + 20 mg/d rosuvastatin	20 mg/d Rosuvastatin						10% [5%]	--	≤0.05
Wiklund 1993 Sweden & Finland [46]	Adults with hypercholesterolemia	12 wk (12 wk)	Pravastatin + gemfibrozil	Placebo	75 (75)	73 (73)	nd	nd	Left the study because of clinical adverse events	4 (5%) [3 (4%)]	1.30 (0.30, 5.60)	nd
				Pravastatin		70 (70)				4 (5%) [2 (3%)]	1.87 (0.35, 9.88)	nd
				Gemfibrozil		72 (72)				4 (5%) [4 (6%)]	0.96 (0.25, 3.69)	nd
Roth 2010 US[47]	Adults with hypercholesterolemia and hypertriglyceridemia	12 wk (8 wk)	5 mg rosuvastatin + fenofibric acid	Simvastatin	118 (118)	119 (119)	nd	22 (16)	AEs leading to discontinuations	6 (5%) [7 (6%)]	0.86 (0.30, 2.50)	nd
			10mg rosuvastatin + fenofibric acid		119 (119)			19 (16)		3 (3%)) [7 (6%)]	0.43 (0.11, 1.62)	nd
			20 mg rosuvastatin + fenofibric acid		118 (118)			23 (16)		4 (3%) [7 (6%)]	0.58 (0.17, 1.92)	nd

[102] All RRs and CIs were calculated by ERT.

Study Year Country Ref #	Population	Duration Outcome measurement (Treatment)	Description		No analyzed (Enrolled)		Baseline GFR or SCr	% with DM	Results			P value
			Intervention	Control	Intervention	Control			Definition	No. (%) Intervention [Control]	RR (95% CI)[102]	
Jones 2010 US[49]	Adults with mixed dyslipidemia	16 wk (12 wk)	Fenofibric acid + atorvastatin and ezetimibe	Placebo + atorvastatin and ezetimibe	272 (272)	270 (271)	nd	20 (23)	AEs leading to discontinuations	16 (6%) [17 (6%)]	0.93 (0.48, 1.81)	nd
Farnier 2010 EU[50]	Adults with mixed dyslipidemia	12 wk (12 wk)	Fenofibrate + pravastatin	Pravastatin	123 (123)	125 (125)	S_{Cr} 84.2 (85.5) µmol/L	26 (29)	Discontinued due to AE	5 (4%) [5 (4%)]	1.02 (0.30, 3.42)	nd
Farnier 2007 Multi[51]	Adults with mixed hyperlipidemia	12 wk (12 wk)	Ezetimibe + simvastatin and fenofibrate	Ezetimibe +, simvastatin	183 (183)	184 (184)	Median S_{Cr} 1.1 (1.1) mg/dL	12 (6)	Discontinued owing to AEs	5 (3%) [5 (3%)]	1.01 (0.30, 3.41)	nd
									Discontinued owing to drug-related AEs	4 (2%) [3 (2%)]	1.34 (0.30, 5.91)	nd
									Discontinued owing to SAEs	0 (0%) [0 (0%)]	--	nd
									Discontinued owing to drug-related SAEs	0 (0%) [0 (0%)]	--	nd
				Fenofibrate		184 (184)	Median S_{Cr} 1.1 (1.0) mg/dL	12 (9)	Discontinued owing to AEs	5 (3%) [6 (3%)]	0.84 (0.26, 2.70)	nd
									Discontinued owing to drug-related AEs	4 (2%) [6 (2%)]	0.67 (0.19, 2.34)	nd
									Discontinued owing to SAEs	0 (0%) [1 (1%)]	--	nd
									Discontinued owing to drug-related SAEs	0 (0%) [1 (1%)]	--	nd

Study Year Country Ref #	Population	Duration Outcome measurement (Treatment)	Description		No analyzed (Enrolled)		Baseline GFR or SCr	% with DM	Results			P value
			Intervention	Control	Intervention	Control			Definition	No. (%) Intervention [Control]	RR (95% CI)[102]	
				Placebo		60 (60)	Median S_{Cr} 1.1 (1.0) mg/dL	12 (10)	Discontinued owing to AEs	5 (3%) [1 (2%)]	1.64 (0.20, 13.76)	nd
									Discontinued owing to drug-related AEs	4 (2%) [0 (0%)]	--	nd
									Discontinued owing to SAEs	0 (0%) [1 (2%)]	--	nd
									Discontinued owing to drug-related SAEs	0 (0%) [0 (0%)]	--	nd
Davidson 2009 US[52]	Adults with dyslipidemia	12 wk (12 wk)	Atorvastatin + fenofibrate	Atorvastatin	73 (73)	74 (74)	S_{Cr} 1.0 (0.9) mg/dL	nd	Total treatment-emergent AEs causing discontinuation	4 (5%) [5 (7%)]	0.81 (0.23, 2.90)	nd
				Fenofibrate		73 (73)				4 (5%) [8 (11%)]	0.50 (0.16, 1.59)	nd
ACCORD 2010 US & Canada[56]	Patients with Type 2 DM at high risk for CVD	5 y (5 y)	Simvastatin +fenofibrate	Simvastatin + placebo	2765 (2765)	2753 (2753)	S_{Cr} 0.9 (0.9) mg/dL	100 (100)	Discontinuation due to a decrease in eGFR	66 (2%) [30 (1%)]	2.19 (1.43, 3.36)	nd
Athyros 2001 Greece[55]	Patients with familial combined hyperlipidemia, 48% with and 52% without CAD at entry	1 y	Pravastatin + gemfibrozil	Atorvastatin	135 (135)	134 (134)	nd	100 (100)	Withdrawn because of side effects	1 (1%) [1 (1%)]	0.99 (0.06, 15.71)	nd
			Simvastatin + gemfibrozil		136 (136)					7 (5%) [1 (1%)]	6.95 (0.87, 55.71)	nd
			Simvastatin + ciprofibrate		134 (134)					3 (2%) [1 (1%)]	2.98 (0.31, 28.27)	nd
			Pravastatin + ciprofibrate		135 (135)					1 (1%) [1 (1%)]	0.99 (0.06, 15.71)	nd

Supplemental Table 25: Patients on statin + fibrate therapy with increased ALT or AST

Study Year Country Ref #	Population	Duration Outcome measurement (Treatment)	Description		No analyzed (Enrolled)		Baseline GFR or SCr	% with DM	Results			P value
			Intervention	Control	Intervention	Control			Definition	No. (%) Intervention [Control]	RR (95% CI)[103]	
Yang 2009 New Zealand[53] [Post hoc analysis of 3 RCTs]	Adults with mixed dyslipidemia	Median 12 wk (Median 12 wk)	Fenofibric acid 135mg + low-dose atorvastatin, rosuvastatin, or simvastatin	Atorvastatin, rosuvastatin, or simvastatin	1895 (1911)		nd	19–26	ALT or AST >3xULN at 2 consecutive visits	0–3% [0–2%]	--	≤0.001
				Fenofibric acid						0–3% [0–4%]	--	NS
Roth 2010 US[47]	Adults with hypercholesterolemia and hypertriglyceridemia	12 wk (8 wk)	5 mg rosuvastatin + fenofibric acid	Simvastatin	117 (118)	114 (119)	nd	22 (16)	ALT >3xULN, consecutive	0 (0%) [0 (0%)]	--	nd
			10mg rosuvastatin + fenofibric acid		118 (119)			19 (16)		1 (1%)) [0 (0%)]	--	nd
			20 mg rosuvastatin + fenofibric acid		115 (118)			23 (16)		2 (2%) [0 (0%)]	--	nd
			5 mg rosuvastatin + fenofibric acid	Simvastatin	117 (118)	114 (119)		22 (16)	ALT >5xULN	2 (2%) [0 (0%)]	--	nd
			10mg rosuvastatin + fenofibric acid		118 (119)			19 (16)		0 (0%) [0 (0%)]	--	nd
			20 mg rosuvastatin + fenofibric acid		115 (118)			23 (16)		1 (1%) [0 (0%)]	--	nd

[103] All RRs and CIs were calculated by ERT.

Study Year Country Ref #	Population	Duration Outcome measurement (Treatment)	Description		No analyzed (Enrolled)		Baseline GFR or SCr	% with DM	Results			P value
			Intervention	Control	Intervention	Control			Definition	No. (%) Intervention [Control]	RR (95% CI)[103]	
			5 mg rosuvastatin + fenofibric acid	Simvastatin	117 (118)	114 (119)		22 (16)	AST >3xULN, consecutive	0 (0%) [0 (0%)]	--	nd
			10mg rosuvastatin + fenofibric acid		118 (119)			19 (16)		0 (0%) [0 (0%)]	--	nd
			20 mg rosuvastatin + fenofibric acid		115 (118)			23 (16)		1 (1%) [0 (0%)]	--	nd
Jones 2010 US & Canada[48] [Subgroup analysis of 3 RCTs]	Adults with mixed dyslipidemia	2 y (2 y)	Fenofibric acid+ low dose rosuvastatin, simvastatin, or atorvastatin	Fenofibric acid	106 (106)	105 (105)	nd	100 (100)	ALT or AST >3xULN, consecutive	2 (2%) [2 (2%)]	0.99 (0.14, 6.90)	nd
				Low dose statin		105 (105)				2 (2%) [0 (0%)]	--	nd
				Moderate dose statin		107 (107)				2 (2%) [0 (0%)]	--	nd
			Fenofibric acid+ moderate dose rosuvastatin, simvastatin, or atorvastatin	Fenofibric acid	110 (110)	105 (105)				1 (1%) [2 (2%)]	0.48 (0.04, 5.19)	nd
				Low dose statin		105 (105)				1 (1%) [0 (0%)]	--	nd
				Moderate dose statin		107 (107)				1 (1%) [0 (0%)]	--	nd

Study Year Country Ref #	Population	Duration Outcome measurement (Treatment)	Description		No analyzed (Enrolled)		Baseline GFR or SCr	% with DM	Results			P value
			Intervention	Control	Intervention	Control			Definition	No. (%) Intervention [Control]	RR (95% CI)[103]	
Jones 2010 US[49]	Adults with mixed dyslipidemia	16 wk (12 wk)	Fenofibric acid + atorvastatin and ezetimibe	Placebo + atorvastatin and ezetimibe	272 (272)	270 (271)	nd	20 (23)	ALT or AST >3xULN, consecutive or >5xULN	14 (5%) [4 (1%)]	3.47 (1.16, 10.42)	NS
Farnier 2010 EU[50]	Adults with mixed dyslipidemia	12 wk (12 wk)	Fenofibrate + pravastatin	Pravastatin	123 (123)	125 (125)	S_{Cr} 84.2 (85.5) µmol/L	26 (29)	Increased AST <3xULN	3 (2%) [0 (0%)]	--	nd
Farnier 2007 Multi[51]	Adults with mixed hyperlipidemia	12 wk (12 wk)	Ezetimibe + simvastatin and fenofibrate	Ezetimibe +, simvastatin	183 (183)	184 (184)	Median S_{Cr} 1.1 (1.1) mg/dL	12 (6)	ALT or AST ≥3x at 2 weekly visits between weeks 11-13	5 (3%) [0 (0%)]	--	nd
				Fenofibrate		184 (184)	Median S_{Cr} 1.1 (1.0) mg/dL	12 (9)		5 (3%) [6 (3%)]	0.84 (0.26, 2.70)	nd
				Placebo		60 (60)	Median S_{Cr} 1.1 (1.0) mg/dL	12 (10)		5 (3%) [0 (0%)]	--	nd

Study Year Country Ref #	Population	Duration Outcome measurement (Treatment)	Description		No analyzed (Enrolled)		Baseline GFR or SCr	% with DM	Results			P value
			Intervention	Control	Intervention	Control			Definition	No. (%) Intervention [Control]	RR (95% CI)[103]	
ACCORD 2010 US & Canada[56]	Patients with Type 2 DM at high risk for CVD	5 y (5 y)	Simvastatin +fenofibrate	Simvastatin + placebo	2765 (2765)	2753 (2753)	S_{Cr} 0.9 (0.9) mg/dL	100 (100)	ALT >3xULN	52 (2%) [40 (1%)]	1.29 (0.86, 1.95)	nd
Athyros 2001 Greece[55]	Patients with familial combined hyperlipidemia, 48% with and 52% without CAD at entry	1 y	Statin + fibrate (pravastatin or simvastatin + gemfibrozil or ciprofibrate)	Atorvastatin	540 (540)	134 (134)	nd	100 (100)	AST>3xULN	7 (1.3%) [0 (0%)]	--	nd

Supplemental Table 26: Patients on statin + fibrate therapy with increased CK

Study Year Country Ref #	Population	Duration Outcome measurement (Treatment)	Description: Intervention	Description: Control	No analyzed (Enrolled): Intervention	No analyzed (Enrolled): Control	Baseline GFR or SCr	% with DM	Results: Definition	Results: No. (%) Intervention [Control]	Results: RR (95% CI)[104]	P value
Yang 2009 New Zealand[53] [Post hoc analysis of 3 RCTs]	Adults with mixed dyslipidemia	Median 12 wk (Median 12 wk)	Fenofibric + low dose atorvastatin, rosuvastatin, or simvastatin	Atorvastatin, rosuvastatin, or simvastatin	1895 (1911)		nd	19–26	CK 5x or 10xULN	0–2% [0–2%]	--	nd
				Fenofibric acid						0–2% [0%]	--	nd
Wiklund 1993 Sweden & Finland[46]	Adults with hypercholesterolemia	12 wk (12 wk)	Pravastatin + gemfibrozil	Placebo	75 (75)	73 (73)	nd	nd	CK increased to 4x baseline at least once	4 (5%) [1 (1%)]	3.89 (0.45, 34.01)	nd
				Pravastatin		70 (70)				4 (5%) [1 (1%)]	3.73 (0.43, 32.60)	nd
				Gemfibrozil		72 (72)				4 (5%) [2 (3%)]	1.92 (0.36, 10.16)	nd
Roth 2010 US[47]	Adults with hypercholesterolemia and hypertriglyceridemia	12 wk (8 wk)	5 mg rosuvastatin + fenofibric acid	Simvastatin	118 (118)	119 (119)	nd	22 (16)	CK >5x ULN	0 (0%) [0 (0%)]	--	nd
			10mg rosuvastatin + fenofibric acid		119 (119)			19 (16)		0 (0%) [0 (0%)]	--	nd
			20 mg rosuvastatin + fenofibric acid		118 (118)			23 (16)		2 (2%) [0 (0%)]	--	nd
			5 mg rosuvastatin + fenofibric acid		118 (118)	119 (119)		22 (16)	CK >10x ULN	0 (0%) [0 (0%)]	--	nd
			10mg rosuvastatin + fenofibric acid		119 (119)			19 (16)		0 (0%) [0 (0%)]	--	nd
			20 mg rosuvastatin + fenofibric acid		118 (118)			23 (16)		1 (1%) [0 (0%)]	--	nd

104 All RRs and CIs were calculated by ERT.

Study Year Country Ref #	Population	Duration Outcome measurement (Treatment)	Description		No analyzed (Enrolled)		Baseline GFR or SCr	% with DM	Results			P value
			Intervention	Control	Intervention	Control			Definition	No. (%) Intervention [Control]	RR (95% CI)[104]	
Jones 2010 US & Canada[48] [Subgroup analysis of 3 RCTs]	Adults with mixed dyslipidemia	2 y (2 y)	Fenofibric acid+ low dose rosuvastatin, simvastatin, or atorvastatin	Fenofibric acid	106 (106)	105 (105)	nd	100 (100)	CK >5x ULN	1 (1%) [0 (0%)]	--	nd
				Low dose statin		105 (105)				1 (1%) [0 (0%)]	--	nd
				Moderate dose statin		107 (107)				1 (1%) [0 (0%)]	--	nd
			Fenofibric acid+ moderate dose rosuvastatin, simvastatin, or atorvastatin	Fenofibric acid	110 (110)	105 (105)				0 (0%) [0 (0%)]	--	nd
				Low dose statin		105 (105)				0 (0%) [0 (0%)]	--	nd
				Moderate dose statin		107 (107)				0 (0%) [0 (0%)]	--	nd
			Fenofibric acid+ low dose rosuvastatin, simvastatin, or atorvastatin	Fenofibric acid	106 (106)	105 (105)			CK >10x ULN	1 (1%) [0 (0%)]	--	nd
				Low dose statin		105 (105)				1 (1%) [0 (0%)]	--	nd
				Moderate dose statin		107 (107)				1 (1%) [0 (0%)]	--	nd
			Fenofibric acid+ moderate dose rosuvastatin, simvastatin, or atorvastatin	Fenofibric acid	110 (110)	105 (105)				0 (0%) [0 (0%)]	--	nd
				Low dose statin		105 (105)				0 (0%) [0 (0%)]	--	nd

Study Year Country Ref #	Population	Duration Outcome measurement (Treatment)	Description		No analyzed (Enrolled)		Baseline GFR or SCr	% with DM	Results			P value
			Intervention	Control	Intervention	Control			Definition	No. (%) Intervention [Control]	RR (95% CI)[104]	
				Moderate dose statin		107 (107)				0 (0%) [0 (0%)]	--	nd
Jones 2010 US[49]	Adults with mixed dyslipidemia	16 wk (12 wk)	Fenofibric acid + atorvastatin and ezetimibe	Placebo + atorvastatin and ezetimibe	272 (272)	270 (271)	nd	20 (23)	CK >10x or >5x ULN	1 (0.3%) [3 (1%)]	0.33 (0.03, 3.16)	NS
Farnier 2010 EU[50]	Adults with mixed dyslipidemia	12 wk (12 wk)	Fenofibrate + pravastatin	Pravastatin	123 (123)	125 (125)	S_{Cr} 84.2 (85.5) μmol/L	26 (29)	CK ≥10xULN with or without muscular symptoms	0 (0%) [1 (1%)]	--	nd
ACCORD 2010 US & Canada[56]	Patients with Type 2 DM at high risk for CVD	5 y (5 y)	Simvastatin +fenofibrate	Simvastatin + placebo	2765 (2765)	2753 (2753)	S_{Cr} 0.9 (0.9) mg/dL	100 (100)	CK >10xULN	10 (0.4%) [9 (0.3%)]	1.11 (0.45, 2.72)	nd

Study Year Country Ref #	Population	Duration Outcome measurement (Treatment)	Description		No analyzed (Enrolled)		Baseline GFR or SCr	% with DM	Results			P value
			Intervention	Control	Intervention	Control			Definition	No. (%) Intervention [Control]	RR (95% CI)[104]	
Athyros 2001 Greece[55]	Patients with familial combined hyperlipidemia, 48% with and 52% without CAD at entry	1 y	Statin + fibrate (pravastatin or simvastatin + gemfibrozil or ciprofibrate)	Atorvastatin	540 (540)	134 (134)	nd	100 (100)	CK >5xULN in simvastatin + gemfibrozil group	3 (1%) [0 (0%)]	--	nd

Supplemental Table 27: Patients on statin + fibrate therapy with increased serum creatinine

Study Year Country Ref #	Population	Duration Outcome measurement (Treatment)	Description		No analyzed (Enrolled)		Baseline GFR or SCr	% with DM	Results			P value
			Intervention	Control	Intervention	Control			Definition	No. (%) Intervention [Control]	RR (95% CI)[105]	
Yang 2009 New Zealand[53] [Post hoc analysis of 3 RCTs]	Adults with mixed dyslipidemia	Median 12 wk (Median 12 wk)	Fenofibric acid 135mg + low-dose atorvastatin, rosuvastatin, or simvastatin	Atorvastatin, rosuvastatin, or simvastatin	1895 (1911)		nd	19–26	↑S_{Cr} ≥50% and >ULN	2–4% [0–2%]	--	<0.05
				Fenofibric acid						2–4% [1–3%]	--	NS
Roth 2010 US[47]	Adults with hypercholesterolemia and hypertriglyceridemia	12 wk (8 wk)	5 mg rosuvastatin + fenofibric acid	Simvastatin	117 (118)	115 (119)	nd	22 (16)	↑S_{Cr} ≥50% and >ULN	1 (1%) [0 (0%)]	--	nd
			10mg rosuvastatin + fenofibric acid		118 (119)			19 (16)		3 (3%) [0 (0%)]	--	nd
			20 mg rosuvastatin + fenofibric acid		115 (118)			23 (16)		2 (2%) [0 (0%)]	--	nd
			5 mg rosuvastatin + fenofibric acid	Simvastatin	117 (118)	115 (119)	nd	22 (16)	↑S_{Cr} ≥100%	0 (0%) [0 (0%)]	--	nd
			10mg rosuvastatin + fenofibric acid		118 (119)			19 (16)		1 (1%) [0 (0%)]	--	nd
			20 mg rosuvastatin + fenofibric acid		115 (118)			23 (16)		0 (0%) [0 (0%)]	--	nd

105 All RRs and CIs were calculated by ERT.

Study Year Country Ref #	Population	Duration Outcome measurement (Treatment)	Description		No analyzed (Enrolled)		Baseline GFR or SCr	% with DM	Results			P value
			Intervention	Control	Intervention	Control			Definition	No. (%) Intervention [Control]	RR (95% CI)[105]	
Jones 2010 US & Canada[48] [Subgroup analysis of 3 RCTs]	Adults with mixed dyslipidemia	2 y (2 y)	Fenofibric acid+ low dose rosuvastatin, simvastatin, or atorvastatin	Fenofibric acid	106 (106)	105 (105)	nd	100 (100)	S_{Cr} >1.5 x baseline and >ULN	4 (4%) [6 (6%)]	0.66 (0.19, 2.27)	nd
				Low dose statin		105 (105)				4 (4%) [2 (2%)]	1.98 (0.37, 110.59)	nd
				Moderate dose statin		107 (107)				4 (4%) [0 (0%)]	--	nd
			Fenofibric acid+ moderate dose rosuvastatin, simvastatin, or atorvastatin	Fenofibric acid	110 (110)	105 (105)				4 (4%) [6 (6%)]	0.64 (0.18, 2.19)	nd
				Low dose statin		105 (105)				4 (4%) [2 (2%)]	1.91 (0.36, 10.20)	nd
				Moderate dose statin		107 (107)				4 (4%) [0 (0%)]	--	nd
			Fenofibric acid+ low dose rosuvastatin, simvastatin, or atorvastatin	Fenofibric acid	106 (106)	105 (105)			S_{Cr} >2 x baseline	1 (1%) [0 (0%)]	--	nd
				Low dose statin		105 (105)				1 (1%) [0 (0%)]	--	nd
				Moderate dose statin		107 (107)				1 (1%) [0 (0%)]	--	nd
			Fenofibric acid+ moderate dose rosuvastatin, simvastatin, or	Fenofibric acid	110 (110)	105 (105)				0 (0%) [0 (0%)]	--	nd
				Low dose statin		105 (105)				0 (0%) [0 (0%)]	--	nd

Study Year Country Ref #	Population	Duration Outcome measurement (Treatment)	Description		No analyzed (Enrolled)		Baseline GFR or SCr	% with DM	Results			P value
			Intervention	Control	Intervention	Control			Definition	No. (%) Intervention [Control]	RR (95% CI)[105]	
			atorvastatin	Moderate dose statin		107 (107)				0 (0%) [0 (0%)]	--	nd
Jones 2010 US[49]	Adults with mixed dyslipidemia	16 wk (12 wk)	Fenofibric acid + atorvastatin and ezetimibe	Placebo + atorvastatin and ezetimibe	267 (272)	267 (271)	nd	20 (23)	↑SCr ≥50% and >1X ULN	1 (0.4%) [1 (0.4%)]	1.00 (0.06, 15.91)	nd
									↑SCr ≥100%	1 (0.4%) [1 (0.4%)]	1.00 (0.06, 15.91)	nd
									SCr > 2 mg/dL	2 (1%) [0 (0%)]	--	nd
Farnier 2010 EU[50]	Adults with mixed dyslipidemia	12 wk (12 wk)	Fenofibrate + pravastatin	Pravastatin	123 (123)	125 (125)	SCr 84.2 (85.5) μmol/L	26 (29)	↓CrCl	3 (2%) [1 (1%)]	3.05 (0.32, 28.91)	nd

Supplemental Table 28: Patients receiving statin + fibrate therapy reporting rhabdomyolysis

Study Year Country Ref #	Population	Duration Outcome measurement (Treatment)	Description		No analyzed (Enrolled)		Baseline GFR or SCr	% with DM	Results			P value
			Intervention	Control	Intervention	Control			Definition	No. (%) Intervention [Control]	RR (95% CI)[106]	
Law 2006 UK&US[57]	Systematic review 1980-2005: 2 cohort studies	1980-2005	Cerivastatin + gemfibrozil	Cerivastatin	nd	nd	nd	nd	Incidence rate of rhadbdomyolysis (95% CI) per 100,000 person-yr	10,300 (3800-22,500) [46 (13-120)]	--	nd
			Any other statin + gemfibrozil	Any other statin						35 (1-194) [3.4 (1.6-6.5)]		
	Same systematic review, 1980-2005, but data are physician reports to FDA Adverse Events Reporting System		Cerivastatin + gemfibrozil	Cerivastatin	nd	nd	nd	nd		15,000 (13,000-17,000) [21 (19-25)]	--	nd
			Any other statin + gemfibrozil	Any other statin						21 (17-25) [0.70 (0.62-0.79)]		
Yang 2009 New Zealand[53] [Post hoc analysis of 3 RCTs]	Adults with mixed dyslipidemia	Median 12 wk (Median 12 wk)	Fenofibric acid+ low-dose atorvastatin, rosuvastatin, or simvastatin	Atorvastatin, rosuvastatin, or simvastatin	1895 (1911)		nd	19-26	Rhabdomyolysis	0 (0%) [0 (0%)]	--	nd
			Fenofibric acid + atorvastatin	Atorvastatin						0 (0%) [0 (0%)]	--	nd
			Fenofibric acid + 10 mg/d rosuvastatin	10 mg/d Rosuvastatin						0 (0%) [0 (0%)]	--	nd

[106] All RRs and CIs were calculated by ERT

Study Year Country Ref #	Population	Duration Outcome measurement (Treatment)	Description		No analyzed (Enrolled)		Baseline GFR or SCr	% with DM	Results			P value
			Intervention	Control	Intervention	Control			Definition	No. (%) Intervention [Control]	RR (95% CI)[106]	
			Fenofibric acid + 20 mg/d rosuvastatin	20 mg/d Rosuvastatin						0 (0%) [0 (0%)]	--	nd
Wiklund 1993 Sweden & Finland[46]	Adults with hypercholesterolemia	12 wk (12 wk)	Pravastatin + gemfibrozil	Placebo	75 (75)	73 (73)	nd	nd	Rhabdomyolysis	0 (0%) [0 (0%)]	--	nd
				Pravastatin		70 (70)				0 (0%) [0 (0%)]	--	nd
				Gemfibrozil		72 (72)				0 (0%) [0 (0%)]	--	nd
Roth 2010 US[47]	Adults with hypercholesterolemia and hypertriglyceridemia	12 wk (8 wk)	5 mg rosuvastatin + fenofibric acid	Simvastatin	118 (118)	119 (119)	nd	22 (16)	Rhabdomyolysis	0 (0%) [0 (0%)]	--	nd
			10 mg rosuvastatin + fenofibric acid		119 (119)			19 (16)		0 (0%) [0 (0%)]	--	nd
			20 mg rosuvastatin + fenofibric acid		118 (118)			23 (16)		0 (0%) [0 (0%)]	--	nd
Jones 2010 US & Canada[48] [Subgroup analysis of 3 RCTs]	Adults with mixed dyslipidemia	2 y (2 y)	Fenofibric acid + low-dose rosuvastatin, simvastatin, or atorvastatin	Fenofibric acid	106 (106)	105 (105)	nd	100 (100)	Rhabdomyolysis	0 (0%) [0 (0%)]	--	nd
				Low-dose statin only		105 (105)				0 (0%) [0 (0%)]	--	nd
				Moderate-dose statin only		107 (107)				0 (0%) [0 (0%)]	--	nd
			Fenofibric acid + moderate-dose rosuvastatin, simvastatin, or atorvastatin	Fenofibric acid	110 (110)	105 (105)				0 (0%) [0 (0%)]	--	nd
				Low-dose statin only		105 (105)				0 (0%) [0 (0%)]	--	nd
				Moderate dose statin only		107 (107)				0 (0%) [0 (0%)]	--	nd
Jones 2010 US[49]	Adults with mixed dyslipidemia	16 wk (12 wk)	Fenofibric acid + atorvastatin and ezetimibe	Placebo + atorvastatin and ezetimibe	272 (272)	270 (271)	nd	20 (23)	Rhabdomyolysis	0 (0%) [0 (0%)]	--	nd

Study Year Country Ref #	Population	Duration Outcome measurement (Treatment)	Description		No analyzed (Enrolled)		Baseline GFR or SCr	% with DM	Results			P value
			Intervention	Control	Intervention	Control			Definition	No. (%) Intervention [Control]	RR (95% CI)[106]	
Farnier 2010 EU [50]	Adults with mixed dyslipidemia	12 wk (12 wk)	Fenofibrate + pravastatin	Pravastatin	123 (123)	125 (125)	SCr 84.2 (8.5) μmol/L	26 (29)	Rhabdomyolysis	0 (0%) [0 (0%)]	--	nd
Farnier 2007 Multi[51]	Adults with mixed hyperlipidemia	12 wk (12 wk)	Ezetimibe + simvastatin and fenofibrate	Ezetimibe + simvastatin	183 (183)	184 (184)	Median SCr 1.1 (1.1) mg/dL	12 (6)	Rhabdomyolysis	0 (0%) [0 (0%)]	--	nd
				Fenofibrate		184 (184)	Median SCr 1.1 (1.0) mg/dL	12 (9)		0 (0%) [0 (0%)]	--	nd
				Placebo		60 (60)	Median SCr 1.1 (1.0) mg/dL	12 (10)		0 (0%) [0 (0%)]	--	nd
Davidson 2009 US[52]	Adults with dyslipidemia	12 wk (12 wk)	Atorvastatin + fenofibrate	Atorvastatin	73 (73)	74 (74)	SCr 1.0 (0.9) mg/dL	nd	Rhabdomyolysis	0 (0%) [0 (0%)]	--	nd
				Fenofibrate		73 (73)	SCr 1.0 (1.0) mg/dL			0 (0%) [0 (0%)]	--	nd

Supplemental Table 29: Summary table of RCTs of statin vs. placebo in kidney transplant patients [categorical outcomes]

Outcome	Author, Year Country Ref #	Duration Outcome measurement (Treatment)	Description: Intervention	Description: Control	No. Analyzed (Enrolled): Intervention	No. Analyzed (Enrolled): Control	Baseline: GFR or S_{Cr}	Baseline: DM (%)	Baseline: TC (mmol/L)	Baseline: LDL (mmol/L)	Baseline: HDL (mmol/L)	Baseline: Tg (mmol/L)	Results: Events No. (%) Intervention [Control]	Results: RR/OR/HR (95% CI)	P value	Quality
Mortality																
All-cause death	ALERT Holdaas 2003 EU & Canada[58]	5 y (5 y)	Fluvastatin	Placebo	1050 (1050)	1052 (1052)	CKD T S_{Cr} 147 (143) μmol/L	19 (19)	6.4 (6.5)	4.1 (4.1)	1.3 (1.4)	2.2 (2.2)	143 (14%) [138 (13%)]	RR 1.02 (0.81, 1.30)	NS	Good
Non-CV deaths													77 (7%) [65 (6%)]	RR 1.20 (0.86, 1.67)	NS	Good
All-cause mortality	ALERT-Extension[107] Holdaas 2005 EU & Canada[59]	7 y (7 y[108])	Fluvastatin	Placebo	819 (819)	833 (833)	CKD T S_{Cr} 144 (139) μmol/L	17 (16)	6.4 (6.5)	4.1 (4.1)	1.3 (1.4)	2.2 (2.2)	194 (19%) [189 (18%)]	RR 1.02 (0.83, 1.24)	NS	Fair
Non-CV death													101 (10%) [90 (9%)]	RR 1.12 (0.84, 1.49)	NS	Fair
CV mortality																
Cardiac death	ALERT Holdaas 2003 EU & Canada[58]	5 y (5 y)	Fluvastatin	Placebo	1050 (1050)	1052 (1052)	CKD T S_{Cr} 147 (143) μmol/L	19 (19)	6.4 (6.5)	4.1 (4.1)	1.3 (1.4)	2.2 (2.2)	36 (3%) [54 (5%)]	RR 0.62 (0.40, 0.96)	0.031	Good
Cardiac death in patients started on statin <4.5 y after transplant	ALERT Holdaas 2005 EU & Canada[60]				522 (522)	521 (521)	CKD T S_{Cr} 144 (142) μmol/L	21 (20)	6.5 (6.4)	4.2 (4.1)	1.3 (1.3)	2.3 (2.3)	nd	RR 0.61 (0.29,1.26)	NS	Good
CV events																
Cardiac death, definite or probable nonfatal MI, CABG/PCI (primary)	ALERT Holdaas 2003 EU & Canada[58]	5 y (5 y)	Fluvastatin	Placebo	1050 (1050)	1052 (1052)	CKD T S_{Cr} 147 (143) μmol/L	19 (19)	6.4 (6.5)	4.1 (4.1)	1.3 (1.4)	2.2 (2.2)	112 (11%) [134 (13%)]	RR 0.83 (0.64, 1.06)	NS	Good
Cardiac death or definite or probable non-fatal MI													79 (8%) [105 (10%)]	RR 0.72 (0.54, 0.97)	0.032	Good

[107] All participants, regardless of the original treatment assignment, received open label fluvastatin.
[108] The patients who completed the core study were offered open-label fluvastatin and active follow-up. Data were collected on those patients who declined active therapy or follow-up at the end of the extension period.

Outcome	Author, Year Country Ref #	Duration Outcome measurement (Treatment)	Description		No. Analyzed (Enrolled)		Baseline						Results		P value	Quality
			Intervention	Control	Intervention	Control	GFR or S_{Cr}	DM (%)	TC (mmol/L)	LDL (mmol/L)	HDL (mmol/L)	Tg (mmol/L)	Events No. (%) Intervention [Control]	RR/OR/HR (95% CI)		
Cardiac death or definite non-fatal MI													70 (7%) [104 (10%)]	RR 0.65 (0.48, 0.88)	0.005 [109]	Good
Definite non-fatal MI													46 (4%) [66 (6%)]	RR 0.68 (0.40, 1.00)	0.050	Good
CABG													25 (2%) [24 (2%)]	RR 1.03 (0.58, 1.81)	NS	Good
PCI													29 (3%) [37 (4%)]	RR 0.80 (0.49, 1.30)	NS	Good
Fatal or non-fatal cerebrovascular events													74 (7%) [63 (6%)]	RR 1.16 (0.83, 1.63)	NS	Good
Cardiac death or non-fatal MI in patients started on statin <4.5 y after transplant	ALERT Holdaas 2005 EU & Canada[60]	5 y (5 y)	Fluvastatin	Placebo	522 (522)	521 (521)	CKD T S_{Cr} 144 (142) μmol/L	21 (20)	6.5 (6.4)	4.2 (4.1)	1.3 (1.3)	2.3 (2.3)	24 (5%) [48 (9%)]	RR 0.44 (0.26,0.74)	0.002	Good
Non-fatal MI in patients started on statin <4.5 y after transplant								16 (18)					nd	RR 0.44 (0.23, 0.84)	0.012	Good
Cardiac death, non-fatal MI, CABG, PCI (primary)	ALERT-Extension[110] Holdaas 2005 EU & Canada[59]	7 y (7 y[111])	Fluvastatin	Placebo	819 (819)	833 (833)	CKD T S_{Cr} 144 (139) μmol/L	17 (16)	6.5 (6.5)	4.1 (4.1)	1.3 (1.4)	2.3 (2.2)	137 (13%) [174 (17%)]	RR 0.79 (0.63, 0.99)	0.036	Fair
Cardiac death, non-fatal MI													95 (9%) [128 (12%)]	RR 0.71 (0.55, 0.93)	0.014	Fair
CABG/PCI													59 (6%) [88 (8%)]	RR 0.67 (0.48, 0.94)	0.019	Fair

109 These data are exploratory and merely show that it is easier to detect a significant reduction in some subgroups than in others. However, the overlapping 95% confidence intervals and negative interaction analysis indicate that there is no significance when individual subgroups (e.g. men and women) are compared with one another. The failure to achieve statistical significance in some subgroups is likely to reflect the low number of events and the low overall power of the study.
110 All participants, regardless of the original treatment assignment, received open label fluvastatin.
111 The patients who completed the core study were offered open-label fluvastatin and active follow-up. Data were collected on those patients who declined active therapy or follow-up at the end of the extension period.

Outcome	Author, Year Country Ref #	Duration Outcome measurement (Treatment)	Description		No. Analyzed (Enrolled)		Baseline						Results		P value	Quality
			Intervention	Control	Intervention	Control	GFR or S_{Cr}	DM (%)	TC (mmol/L)	LDL (mmol/L)	HDL (mmol/L)	Tg (mmol/L)	Events No. (%) Intervention [Control]	RR/OR/HR (95% CI)		
Fatal or non-fatal cerebrovascular events													93 (9%) [91 (9%)]	RR 1.01 (0.75, 1.35)	NS	Fair
CV event[112] in patients with SLE (primary)	ALERT-Extension Norby 2009. EU & Canada[61]	7 y (7 y[113])	Fluvastatin	Placebo	23 (23)	10 (10)	S_{Cr} 133.5 (129.4) μmol/L	nd	6.4 (6.2)	4.0 (3.9)	1.4 (1.3)	2.2 (1.9)	3 (13%) [4 (40%)]	RR 26.6 (5.9, 119.4)	NS (0.064)	Good
Cardiac death or definite MI in patients with DM	ALERT Jardine 2004 EU & Canada[62]	5 y (5 y)	Fluvastatin	Placebo	nd	nd	nd	100 (100)	nd	4.1 (4.1)	nd	nd	25 (13%) [34 (17%)]	RR 0.71 (0.41, 1.21)	nd	Fair
Cardiac death or definite MI in patients without DM								0 (0)					45 (5%) [70 (8%)]	RR 0.62 (0.43, 0.91)	nd	Fair
Cardiac death or definite MI in patients with CHD								19 (19)					20 (20%) [24 (24%)]	RR 0.72 (0.39, 1.34)	nd	Fair
Cardiac death or definite MI in patients without CHD								19 (19)					50 (5%) [80 (8%)]	RR 0.63 (0.44, 0.89)	nd	Fair
Graft loss																
Graft loss or doubling of SCr	ALERT Holdaas 2003 EU & Canada[58]	5 y (5 y)	Fluvastatin	Placebo	1050 (1050)	1052 (1052)	CKD T S_{Cr} 147 (143) μmol/L	19 (19)	6.4 (6.5)	4.1 (4.1)	1.3 (1.4)	2.2 (2.2)	183 (17%) [165 (16%)]	RR 1.10 (0.89, 1.36)	NS	Good
Graft loss													146 (14%) [137 (13%)]	RR 1.005 (0.835, 1.334)[114]	NS	Good
Graft loss or doubling of SCr or death													276 (26%) [268 (26%)]	RR 1.02 (0.86, 1.21)	NS	Good

[112] Nonfatal MI cardiac death and coronary intervention procedure
[113] The patients who completed the core study were offered open-label fluvastatin and active follow-up. Data were collected on those patients who declined active therapy or follow-up at the end of the extension period.
[114] Calculated by ERT.

Outcome	Author, Year Country Ref #	Duration Outcome measurement (Treatment)	Description		No. Analyzed (Enrolled)		Baseline						Results		P value	Quality
			Intervention	Control	Intervention	Control	GFR or S_{Cr}	DM (%)	TC	LDL	HDL	Tg	Events No. (%) Intervention [Control]	RR/OR/HR (95% CI)		
									(mmol/L)							
Graft loss in patients with moderate renal impairment	ALERT Fellstrom 2006 EU & Canada[63]	5 y (5 y)	Fluvastatin	Placebo	318 (318)	336 (336)	GFR<50 mL/min	19 (19)	nd	nd	nd	nd	91 (29%) [87 (26%)]	RR 1.11 (0.86, 1.42)	NS	Good
Graft loss in patients with severe renal impairment					45 (45)	49 (49)	GFR<30 mL/min						28 (62%) [25 (51%)]	RR 1.22 (0.85, 1.74)	NS	Good
Graft loss in all patients	ALERT Fellstrom 2006 EU & Canada[63]	5 y (5 y)	Fluvastatin	Placebo	1050 (1050)	1052 (1052)	SCr 143 µmol/L	19 (19)	6.5	4.1	1.4	2.2	146 (14%) 137 (13%)	RR 1.07 (0.86, 1.33)	NS	Good
First treated acute rejection (primary)	Holdaas 2001 EU[64]	3 mo (3 mo)	Fluvastatin	Placebo	182 (182)	182 (182)	CKD T SCr 160 umol/L	12 (14)	4.77 (4.74)	2.99 (2.96)	1.02 (1.02)	1.68 (1.55)	86 (47%) [87 (48%)]	RR 0.99 (0.80, 1.23)	NS	Good
First steroid-resistant acute rejection													38 (21%) [34 (19%)]	RR 1.12 (0.74, 1.69)	NS	Good
First biopsy-confirmed acute rejection													70 (39%) [75 (41%)]	RR 0.93 (0.72, 1.20)	NS	Good
Days to treated first acute rejection													19 [18]	--	NS	Good
Second acute rejection episode													18 (10%) [19 (10%)]	RR 0.95 (0.51, 1.75)	NS	Good

Outcome	Author, Year Country Ref #	Duration Outcome measurement (Treatment)	Description		No. Analyzed (Enrolled)		Baseline						Results		P value	Quality
			Intervention	Control	Intervention	Control	GFR or S_{Cr}	DM (%)	TC	LDL	HDL	Tg	Events No. (%) Intervention [Control]	RR/OR/HR (95% CI)		
									(mmol/L)							
Graft loss or death													12 (6.6%) [7 (3.8%)]	RR 1.71 (0.69, 4.26)	NS	Good
Biopsy-confirmed rejection, graft loss, or death													77 (42%) [80 [44%])	RR 0.96 (0.76, 1.22)	NS	Good
Severity of rejection-(BANFF I) Mild													26 (14%) [42 (23%)]	--		Good
Severity of rejection-(BANFF II) Moderate													36 (20%) [27 (15%)]	--	NS (0.091)	Good
Severity of rejection-(BANFF III) Severe													8 (4%) [6 (3%)]	--		Good

Supplemental Table 30: Summary table of RCTs of statin vs. placebo in kidney transplant patients [continuous outcomes]

Outcome	Author, Year Country Ref #	Duration Outcome measurement (Treatment)	Description		No. Analyzed (Enrolled)		Baseline		Results (Lipids)				P value	Quality
			Intervention	Control	Intervention	Control	GFR or S_{Cr}	DM (%)	Baseline Intervention (Control)	Final Intervention (Control)	Δ Intervention (Control)	Net Δ (95% CI)		
Lipid levels														
ΔLDL, mmol/L[115]	ALERT Holdaas 2003 EU & Canada[58]	6 wk (5 y)	Fluvastatin	Placebo	1050 (1050)	1052 (1052)	CKD T S_{Cr} 147 (143) μmol/L	19 (19)	4.1 (4.1)	3.1[116] (4.0)	-1.0 (-0.1)	-0.9	nd	Good
		5 y (5 y)								2.8[117] (3.8)	-1.3 (-0.3)	-1.0	nd	Good
ΔTotal cholesterol, mmol/L		6 wk (5 y)							6.4 (6.5)	5.2[118] (6.4)	-1.3 (-0.1)	-1.2	nd	Good
		5 y (5 y)								4.9[119] (6.0)	-1.5 (-0.5)	-1.0	nd	Good
ΔLDL in patients with SLE, mmol/L	ALERT Norby 2009 EU & Canada[61]	7 y (7 y[120])	Fluvastatin	Placebo	23 (23)	10 (10)	S_{Cr} 133.5 (129.4) μmol/L	nd	4.0 (3.9)	2.8 (nd)	29.2% (18.3, 40)	--	nd	Fair
ΔTotal cholesterol in patients with SLE, mmol/L									6.4 (6.2)	5.1 (nd)	19.6% (11.7, 27.5)	--	nd	Fair
Total cholesterol, mmol/L	Holdaas 2001 EU[64]	3 mo (3 mo)	Fluvastatin	Placebo	nd (182)	nd (182)	CKD T	12 (14)	4.77 (4.74)	5.47 (6.08)	0.70 (1.34)	−0.73 (-1.08, -0.37)	<0.001	Good
LDL cholesterol, mmol/L									2.99 (2.96)	3.06 (3.74)	0.07 (0.78)	−0.84 (−1.12, −0.56)	<0.001	Good
HDL cholesterol, mmol/L									1.02 (1.02)	1.40 (1.32)	0.38 (0.30)	0.08 (−0.03, 0.19)	NS	Good
Triglycerides, mmol/L									1.68 (1.55)	2.04 (2.24)	0.36 (0.69)	−0.30 (−0.57, −0.02)	0.034	Good

[115] During the course of the study, 696 (66.3%) of the patients in the fluvastatin group achieved total cholesterol concentrations lower than 5·0 mmol/L, and 779 (74·2%) achieved an LDL cholesterol concentrations lower than 3.0 mmol/L, meeting recommended values for prevention of CHD.
[116] Estimated from figure
[117] Estimated from figure
[118] Estimated from figure
[119] Estimated from figure
[120] The patients who completed the core study were offered open-label fluvastatin and active follow-up. Data were collected on those patients who declined active therapy or follow-up at the end of the extension period.

Supplemental Table 31: Evidence profile of RCTs examining the effect of statins vs. placebo in kidney transplant recipients

Outcome	# of studies and study design	Total N (treatment)	Methodological quality of studies per outcome	Consistency across studies	Directness of the evidence generalizability/ applicability	Other considerations	Summary of findings		
							Quality of evidence for outcome	Qualitative and quantitative description of effect	Importance of outcome
Mortality	1 RCT (High)	2102 (1050)	No limitations (0)	NA	Direct (0)	Sparse (-1)	Moderate	No difference	Critical
CV mortality	1 RCT (High)	2102 (1050)	No limitations (0)	NA	Direct (0)	Sparse (-1)	Moderate	Benefit from statins	Critical
CV events	1 RCT (High)	2102 (1050)	No limitations (0)	NA	Direct (0)	Sparse (-1)	Moderate	Benefit from statins	Critical
ESRD	0 RCTs	--	--	--	--	--	--	--	Critical
Graft loss	2 RCTs (High)	2466 (1232)	No limitations (0)	No important inconsistencies (0)	Direct (0)	None (0)	High	No difference	High
Kidney function (categorical)	0 RCTs	--	--	--	--	--	--	--	High
Lipid levels (continuous)	2 RCTs (High)	2466 (1232)	No limitations (0)	Important inconsistencies (-1)	Uncertainty about directness (-1)	None (0)	Low	Benefit from statins	Moderate
Adverse events	2 RCTs (High)	2466 (1232)						No difference	Moderate
Total	2 RCTs (High)	2466 (1232)							

Balance of potential benefits and harms:
Benefit for CV events

Quality of overall evidence:
Moderate

Supplemental Table 32: Summary table of RCTs of statins vs. placebo in children with CKD without DM [continuous outcomes]

Outcome	Author, Year Country Ref #	Duration Outcome measurement (Treatment)	Description		No. Analyzed (Enrolled)		Baseline		Results (Lipids)				P value	Quality
			Intervention	Control	Intervention	Control	GFR or S_{Cr}	DM (%)	Baseline Intervention (Control)	Final Intervention (Control)	Δ Intervention (Control)	Net Δ (95% CI)		
Lipid levels														
ΔTotal cholesterol, mmol/L	Mackie 2010[65] Australia	2 mo (2 mo)	Atorvastatin 10 mg/d	Placebo	8 (8)	8 (8)	S_{Cr} 293 (284) μmol/L	nd	5.1 (4.9)	3.8 (5.0)	-1.3 (0.1)	-1.2	<0.001	Fair
ΔLDL, mmol/L									3.1 (3.0)	1.9 (3.1)	-1.1 (0.2)	-0.9	<0.001	Fair
ΔTriglycerides, mmol/L									1.2 (1.2)	1.2 (1.5)	-0.4 (+0.3)	-0.1	nd	Poor

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